



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

ASSIGNMENT - 1

The domain of the Project:
Physical Design (VLSI)

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TASK - A

1. Basic Linux Commands

Command	Purpose	Syntax	Description
ls	List directory contents	ls, ls -l, ls -a	Displays files and folders in current directory. -l gives detailed view, -a shows hidden files.
pwd	Print working directory	pwd	Shows full path of current directory.
mkdir	Create directory	mkdir folder_name	Creates a new directory.
rmdir	Remove empty directory	rmdir folder_name	Deletes an empty directory only.
cd	Change directory	cd dir_name, cd .., cd ~	Used to navigate between directories.
gedit	Graphical text editor	gedit file.txt	Opens file in GUI editor for editing.
vi	Terminal text editor	vi file.txt	Command-line editor with insert and command modes.
cp	Copy files/directories	cp source destination, cp - r dir1 dir2	Copies files or folders.
mv	Move or rename files	mv old new, mv file path/	Moves or renames files/directories.



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rm	Remove files/directories	rm file, rm -r folder, rm -f file	Deletes files or directories (use carefully).
diff (vi diff)	Compare files	diff file1 file2	Shows line-by-line differences between two files.
man	Manual pages	Man command_name	Displays detailed help and usage of commands.

2. Vi commands

Basically it is used for edit the file the commands are used for this to save the edits

1. vi <file name > to open the file or create the file
2. :i to edit the file
3. :w to save the file
4. :q to quit from the file
5. :wq to save & quit from the file
6. :wq! to save and quit forcefully from the file
7. h, j, k, l movement [left, down ,up ,right]
8. ESC key to exit from command window
9. u undo the file contents
10. x delete the file contents single character
11. dw delete the word in the file
12. dd delete the entire contents of file
13. :set nu to set point line number
14. :set nonu to remove point line number
15. gg to move to start line
16. G move to last line
17. /<text> Search forward for a word (press n to find the next occurrence).

3. Chmod (Change Permissions):

The chmod command in Linux is used to change file or directory permissions. It controls who can read, write, or execute a file.

In numeric mode, permissions are represented using **3 digits**, where each digit ranges from **0 to 7**. Each digit represents a user category: **Owner, Group, and Others**. The values are based on:

- Read = 4
- Write = 2
- Execute = 1

So, combinations form values like 7 (rwx), 6 (rw-), 5 (r-x), and 4 (r--).



4. grep (Search Pattern in Files):

The grep command is used to search for specific text patterns inside files. It scans each line and displays only the lines that match the given pattern. It is widely used in log analysis, debugging, and data filtering.

Syntax: `grep "pattern" filename` (grep "error" log.txt, grep "hello" file.txt)

5. awk (Text Processing Tool):

The awk command is a powerful text processing tool used for scanning, extracting, and manipulating structured data. It processes input line by line and divides each line into fields, making it useful for report generation and data analysis.

Syntax: `awk 'pattern {action}' filename` (awk '{print \$1}' data.txt, awk '{print \$1, \$3}' file.txt)

List of AWK Commands

`awk '{print}' file.txt` → Print entire file

`awk '{print $1}' file.txt` → Print first column

`awk '{print $2}' file.txt` → Print second column

`awk '{print $1, $3}' file.txt` → Print selected columns

`awk '/pattern/ {print}' file.txt` → Search and print matching lines

`awk 'NR==1 {print}' file.txt` → Print first line only

`awk 'NR>1 {print}' file.txt` → Skip first line

`awk '{print NR, $0}' file.txt` → Print line numbers with content

`awk 'length($0)>10' file.txt` → Print lines with length > 10

`awk '{sum+=$1} END {print sum}' file.txt` → Sum first column values

`awk '{print NF}' file.txt` → Print number of fields per line

`awk 'END {print NR}' file.txt` → Count total number of lines

TASK - B

Open Lane Environment:

OpenLane is an open-source VLSI Physical Design automation flow used for converting RTL designs into GDSII layout. It integrates multiple tools such as synthesis, floorplanning, placement, routing, and verification. Understanding its directory structure and logs is essential for debugging and successful chip design execution.



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The Open Lane environment consists of multiple important directories:

Directory/File	Purpose
designs/	Contains RTL designs, config files, and run folders
scripts/	Contains TCL and automation scripts used in the flow
flow.tcl	Main TCL script used to control the OpenLane flow
runs/	Stores run-specific logs, reports, and generated results
pdks/	Contains technology-specific Process Design Kit files
results/	Stores generated DEF, GDSII, netlists, and layout outputs
logs/	Stores logs for synthesis, placement, CTS, and routing stages

The OpenLane flow is executed using TCL commands through the flow.tcl script. During execution, separate logs, reports, and result files are generated for each stage such as synthesis, floorplanning, placement, CTS, and routing. These files help in debugging and analyzing the design.

Commands Used to Launch OpenLane

```
docker run -it -v $PWD:/openlane efabless/openlane:2023.09.07
```

Starting Interactive Mode

```
cd OpenLane
./flow.tcl -interactive
```

Preparing Design

```
prep -design spm
```

```
penLane Container:/openlane% cd designs
penLane Container:/openlane/designs% la
sh: command not found: la
penLane Container:/openlane/designs% ls
t  spm
penLane Container:/openlane/designs% cd spM
d: no such file or directory: spM
penLane Container:/openlane/designs% cd spm
penLane Container:/openlane/designs/spm% ls
onfig.json  pin_order.cfg  runs  src  verify
penLane Container:/openlane/designs/spm% cd runs
penLane Container:/openlane/designs/spm/runs% ls
UN_2026.05.05_17.48.20  RUN_2026.05.05_18.11.36  RUN_2026.05.05_22.43.48  RUN_2026.05.05_22.49.30
penLane Container:/openlane/designs/spm/runs% cd RUN_
d: no such file or directory: RUN_
penLane Container:/openlane/designs/spm/runs% cd RUN_2026.05.05_18.11.36
penLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36% ls
PENLANE_VERSION  cmd5.log  config_in.tcl  openlane.log  results  tmp
DK_SOURCES       config.tcl  logs          reports        runtime.yaml  warnings.log
penLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36% cd reports
penLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports% cd synthesis
penLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/synthesis% ls
-synthesis.AREA_0.chk.rpt  1-synthesis_pre.stat  2-syn_sta.max.rpt  2-syn_sta.skew.rpt
-synthesis.AREA_0.stat.rpt  1-synthesis_pre_synth.chk.rpt  2-syn_sta.min.rpt  2-syn_sta.summary.rpt
-synthesis_dff.stat  2-syn_sta.checks.rpt  2-syn_sta.power.rpt
penLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/synthesis% cd ..
```



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```
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/synthesis% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports% cd floorplan
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/floorplan% ls
3-initial_fp_core_area.rpt 3-initial_fp_die_area.rpt
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/floorplan% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports% cd placement
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/placement% ls
11-dpl_sta.checks.rpt 11-dpl_sta.power.rpt 8-gpl_sta.checks.rpt 8-gpl_sta.power.rpt
11-dpl_sta.max.rpt 11-dpl_sta.skew.rpt 8-gpl_sta.max.rpt 8-gpl_sta.skew.rpt
11-dpl_sta.min.rpt 11-dpl_sta.summary.rpt 8-gpl_sta.min.rpt 8-gpl_sta.summary.rpt
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/placement% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports% cd routing
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/routing% ls
16-rsz_design_sta.checks.rpt 18-rsz_timing_sta.checks.rpt 21-grt_sta.checks.rpt 23-drt_metrics.json
16-rsz_design_sta.max.rpt 18-rsz_timing_sta.max.rpt 21-grt_sta.max.rpt 24-wire_lengths.csv
16-rsz_design_sta.min.rpt 18-rsz_timing_sta.min.rpt 21-grt_sta.min.rpt drt.drc
16-rsz_design_sta.power.rpt 18-rsz_timing_sta.power.rpt 21-grt_sta.power.rpt drt.klayout.xml
16-rsz_design_sta.skew.rpt 18-rsz_timing_sta.skew.rpt 21-grt_sta.skew.rpt
16-rsz_design_sta.summary.rpt 18-rsz_timing_sta.summary.rpt 21-grt_sta.summary.rpt
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/routing% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports% cd signoff
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff% ls
26-sta-rcx_min 31-irdrop-VGND.rpt 34-xor.xml 41-antenna_violators.rpt drc.rdb drc.tr
28-sta-rcx_max 31-irdrop-VPWR.rpt 35-spice.ext_feedback.txt 41-antenna_violators_pins.txt drc.rpt violations.json
30-sta-rcx_nom 34-xor.rpt 38-spm.lvs.rpt drc.klayout.xml drc.tcl violations.xml
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff% cd ../../results/final
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final% ls
def gds lef lib mag maglef sdc sdf spef spi verilog
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final% cd def
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/def% ls
spm.def
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/def% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/lef% ls
spm.lef
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/lef% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final% cd lib
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/lib% ls
spm.lib
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/lib% cd ..
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final% cd spef
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/spef% ls
multicorner spm.spef
OpenLane Container:/openlane/designs/spm/runs/RUN_2026.05.05_18.11.36/results/final/spef% █
```

Fig1.1. Exploring the Directory using cd and ls commands

TASK – C

TCL Commands Used in OpenLane

```
prep -design spm
run_synthesis
run_floorplan
run_placement
run_cts
run_routing
```

The above TCL commands were executed in OpenLane interactive mode to perform the ASIC physical design flow.

TCL:

1. Wire Length Calculation

- Find out total Wire length, Clock wire length and Data wire length of design



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Method 1: Using Basic commands from 25-wire_lengths.csv

```
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_13.46.46/reports/routing$ ls
25-wire_lengths.csv  drc.drc  drc.klayout.xml
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_13.46.46/reports/routing$ awk -F',' '{sum += $2}
END {print sum}' 25-wire_lengths.csv
6326.73
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_13.46.46/reports/routing$ grep -i clk 25-wire_len
gths.csv | awk -F',' '{sum += $2} END {print sum}'
779.37
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_13.46.46/reports/routing$ grep -iv clk 25-wire_le
ngths.csv | awk -F',' '{sum += $2} END {print sum}'
5547.36
```

a. Total wire length: 6326.73

awk -F',' '{sum += \$2} END {print sum}' 25-wire_lengths.csv

b. Clock wire length: 779.37

grep -i clk 25-wire_lengths.csv | awk -F',' '{sum += \$2} END {print sum}'

c. Data wire length: 5547.36

grep -iv clk 25-wire_lengths.csv | awk -F',' '{sum += \$2} END {print sum}'

Method 2- Using Openroad and Manual calculations

```
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ head reports/routing/24-wire_lengths.cs
v
net,length_um
_097_,173.32
_165_,145.95
clknet_0_clk,145.53
_122_,139.855
_099_,138.2
_133_,131.66
_205_,127.38
_201_,127.25
_154_,124.03
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {printf "
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {printf "
%.0f\n", sum*1000000000}' reports/routing/24-wire_lengths.csv
6716815000000
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {print "T
otal Wire Length =", sum, "um"}' reports/routing/24-wire_lengths.csv
Total Wire Length = 6716.82 um
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {printf "
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {printf "
%.0f\n", sum*1000000000}' reports/routing/24-wire_lengths.csv
6716815000000
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {print "T
otal Wire Length =", sum, "um"}' reports/routing/24-wire_lengths.csv
Total Wire Length = 6716.82 um
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, 'NR>1 {sum += $2} END {print "T
otal Wire Length =", sum, "um"}' reports/routing/24-wire_lengths.csv
Total Wire Length = 6716.82 um
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, '/clk/ {sum += $2} END {print "
Clock Wire Length =", sum, "um"}' reports/routing/24-wire_lengths.csv
Clock Wire Length = 784.48 um
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ awk -F, '
> NR>1 {total += $2}
> /clk/ {clk += $2}
> END {print "Data Wire Length =", total-clk, "um"}
> ' reports/routing/24-wire_lengths.csv
Data Wire Length = 5932.34 um
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$
```



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After opening the openroad and executing these commands

- run_synthesis
- run_floorplan
- run_placement
- run_cts
- run_routing

We get the total wire length of design as: 6716.82 um

Clock wire length will be approximately 10% of total wire length: $0.1 * 6716.82 = 671.682$ um

Data wire length = Wire length - Clock wire: $6716.82 - 671.682 = 6045.138$ um

2. Find the worst IR drop and the % degradation

The script to run is:

- cd /openlane/designs/spm/runs/openlane_test/reports/signoff
- ls • ls | grep -i "pwr"
- head -20 32-irdrop-VPWR.rpt
- \$awk -F',' '{print \$NF}' 32-irdrop-VPWR.rpt | head

```
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ cd /openlane/designs/spm/runs/RUN_2026.05.07_15.10.44
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ sort -k4 -n reports/signoff/32-irdrop-VPWR.rpt | head -10
Instance name, X location, Y location, Voltage
_229_, 5.52, 35.36, 1.79947
_451_, 5.52, 35.36, 1.79947
clkbuf_3_0_f_clk, 5.52, 35.36, 1.79947
PHY_16, 5.52, 35.36, 1.79947
PHY_18, 5.52, 35.36, 1.79947
_225_, 28.172, 40.8, 1.79955
_467_, 28.172, 40.8, 1.79955
clkbuf_3_1_f_clk, 28.172, 40.8, 1.79955
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ echo "1.8 - 1.79947" | bc -l
.00053
OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.07_15.10.44$ echo "(0.00053/1.8)*100" | bc -l
.02944444444444444400
```




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```
# ----- OUTPUT -----
puts "\n===== IR DROP REPORT ====="
puts "VDD MIN VOLTAGE : $vdd_min V"
puts "VDD IR DROP      : $vdd_ir_drop V"
puts "VDD % DROP        : $vdd_percent %"
puts "-----"
puts "VSS MAX VOLTAGE : $vss_max V"
puts "===== % designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff/31-irdrop-VPWR.rpt
% designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff/31-irdrop-VGND.rpt
% % % 999
% 0
% % % file3
% 53
% % % % % file3
% 53
% % % % % 0.0001320000000000021
% 0.000093
% % 0.0073333333333334499
% % %
===== IR DROP REPORT =====
% VDD MIN VOLTAGE : 1.799868 V
% VDD IR DROP      : 0.0001320000000000021 V
% VDD % DROP        : 0.0073333333333334499 %
% -----
% VSS MAX VOLTAGE : 0.000093 V

OpenLane Container:/openlane% cd /openlane
OpenLane Container:/openlane% head designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff/31-irdrop-VGND.rpt
Instance,Terminal,Layer,X location,Y location,Voltage
FILLER_0_30_178,VGND,li1,87.8600,92.4800,0.000019
FILLER_0_30_172,VGND,li1,85.3300,92.4800,0.000019
FILLER_0_30_166,VGND,li1,82.3400,92.4800,0.000018
FILLER_0_30_160,VGND,li1,79.8100,92.4800,0.000017
FILLER_0_30_154,VGND,li1,77.0500,92.4800,0.000015
FILLER_0_30_148,VGND,li1,74.2900,92.4800,0.000013
FILLER_0_30_144,VGND,li1,71.9900,92.4800,0.000014
FILLER_0_30_136,VGND,li1,69.0000,92.4800,0.000016
FILLER_0_30_130,VGND,li1,66.0100,92.4800,0.000018
OpenLane Container:/openlane% head designs/spm/runs/RUN_2026.05.05_18.11.36/reports/signoff/31-irdrop-VPWR.rpt
Instance,Terminal,Layer,X location,Y location,Voltage
FILLER_0_30_178,VPWR,li1,87.8600,95.2000,1.800000
FILLER_0_30_172,VPWR,li1,85.3300,95.2000,1.800000
FILLER_0_30_166,VPWR,li1,82.3400,95.2000,1.800000
FILLER_0_30_160,VPWR,li1,79.8100,95.2000,1.800000
FILLER_0_30_154,VPWR,li1,77.0500,95.2000,1.800000
FILLER_0_30_148,VPWR,li1,74.2900,95.2000,1.800000
FILLER_0_30_144,VPWR,li1,71.9900,95.2000,1.800000
FILLER_0_30_136,VPWR,li1,69.0000,95.2000,1.800000
FILLER_0_30_130,VPWR,li1,66.0100,95.2000,1.800000
```

The IR drop report provides voltage values across different instances in the design. The minimum voltage value corresponds to the worst IR drop location.

- From observation the lowest voltage is 1.79972 V
- IR Drop can be calculated using the formula: Actual voltage (VDD) – Lowest voltage
- Sky130 technology uses 1.8 V as nominal supply voltage. This is a standard defined by the PDK, not in design config.
- Then the worst IR drop is: $1.8 \text{ V} - 1.79972 = 0.00028 \text{ V}$
- To find Percentage Degradation we use: $(\text{IR Drop}/\text{VDD}) * 100$
- Then, the percentage Degradation is $(0.00028/1.8) * 100 = 0.01555555\%$

3. Find which cell receives less voltage? Top 10

a. Needs both IR drop of VDD and VSS



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The script used to get the top 10 cells receiving less voltage including both VDD and VSS:

- cd Openlane/OpenLane
- make mount
- cd /openlane/designs/spm/runs/openlane_test/reports/signoff
- ls | grep -i "pwr"
- ls | grep -i "VGND"
- head -20 32-irdrop-VPWR.rpt
- head -20 32-irdrop-VGND.rpt

```
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
_327_, 84.81, 89.76, 1.79971
_424_, 84.81, 89.76, 1.79971
_492_, 84.81, 89.76, 1.79971
FILLER_0_28_161, 84.81, 89.76, 1.79971
FILLER_0_29_181, 84.81, 89.76, 1.79971
FILLER_0_30_173, 84.81, 89.76, 1.79971
_427_, 84.81, 84.32, 1.79973
_491_, 84.81, 84.32, 1.79973
FILLER_0_27_169, 84.81, 84.32, 1.79973
_333_, 84.81, 78.88, 1.79974
_426_, 84.81, 78.88, 1.79974
FILLER_0_25_169, 84.81, 78.88, 1.79974
_337_, 84.81, 73.44, 1.79977
_340_, 84.81, 73.44, 1.79977
FILLER_0_22_171, 84.81, 73.44, 1.79977
FILLER_0_23_169, 84.81, 73.44, 1.79977
FILLER_0_24_183, 84.81, 73.44, 1.79977
_338_, 84.81, 68, 1.79976
_341_, 84.81, 68, 1.79976
_494_, 84.81, 68, 1.79976
hold45, 84.81, 68, 1.79976
reports/signoff/32-irdrop-VPWR.rpt

sanjana-s@sanjana-s-VirtualBox: ~/OpenLane
Instance name, X location, Y location, Voltage
_329_, 88.11, 97.92, 0.000368534
_489_, 88.11, 97.92, 0.000368534
input21, 88.11, 97.92, 0.000368534
input22, 88.11, 97.92, 0.000368534
input23, 88.11, 97.92, 0.000368534
input25, 88.11, 97.92, 0.000368534
_327_, 88.11, 92.48, 0.000375223
_424_, 88.11, 92.48, 0.000375223
_425_, 88.11, 92.48, 0.000375223
hold50, 88.11, 92.48, 0.000375223
FILLER_0_30_173, 88.11, 92.48, 0.000375223
_427_, 88.11, 87.04, 0.000356275
_492_, 88.11, 87.04, 0.000356275
FILLER_0_27_169, 88.11, 87.04, 0.000356275
FILLER_0_29_181, 88.11, 87.04, 0.000356275
_426_, 88.11, 81.6, 0.000359686
hold48, 88.11, 81.6, 0.000359686
FILLER_0_25_169, 88.11, 81.6, 0.000359686
_332_, 88.11, 76.16, 0.000349487
_333_, 88.11, 76.16, 0.000349487
_337_, 88.11, 76.16, 0.000349487
FILLER_0_23_169, 88.11, 76.16, 0.000349487
FILLER_0_24_183, 88.11, 76.16, 0.000349487
_494_, 88.11, 70.72, 0.000351047
hold45, 88.11, 70.72, 0.000351047
FILLER_0_21_169, 88.11, 70.72, 0.000351047
_338_, 88.11, 65.28, 0.000345893
_341_, 88.11, 65.28, 0.000345893
_430_, 88.11, 65.28, 0.000345893
reports/signoff/32-irdrop-VGND.rpt
```

The IR-drop analysis shows that the design has very small voltage degradation on both VPWR and VGND rails. The minimum VPWR observed is approximately 1.79971 V, while the maximum VGND bounce observed is approximately 0.000356275 V. Hence the power distribution network is stable and reliable.



4. Levels of logic

```
OpenLane Container (ff5589f):/openlane/designs/spm/runs/RUN_2026.05.23_12.49.50/reports/routing$ grep -iE "start|end|slack" 21-grt_sta.max.rpt |head -n 60
Startpoint: _444_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: p (output port clocked by clk)
6.86 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _467_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _471_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _473_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _481_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _465_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _469_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _479_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _483_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _503_ (rising edge-triggered flip-flop clocked by clk)
7.17 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _477_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _475_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _487_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _447_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _451_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _501_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _443_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _495_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
Startpoint: y (input port clocked by clk)
Endpoint: _497_ (rising edge-triggered flip-flop clocked by clk)
7.18 slack (MET)
```

Path #	Start point (SP)	End point (EP)	Slack Value	LOL
1	444 (Flip-Flop)	p (Output Port)	6.86 slack (MET)	1
2	y (Input Port)	467 (Flip-Flop)	7.17 slack (MET)	5
3	y (Input Port)	471 (Flip-Flop)	7.17 slack (MET)	5
4	y (Input Port)	473 (Flip-Flop)	7.17 slack (MET)	5
5	y (Input Port)	481 (Flip-Flop)	7.17 slack (MET)	5
6	y (Input Port)	465 (Flip-Flop)	7.17 slack (MET)	5
7	y (Input Port)	469 (Flip-Flop)	7.17 slack (MET)	5
8	y (Input Port)	479 (Flip-Flop)	7.17 slack (MET)	5
9	y (Input Port)	483 (Flip-Flop)	7.17 slack (MET)	5
10	y (Input Port)	503 (Flip-Flop)	7.18 slack (MET)	5



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11	y (Input Port)	477 (Flip-Flop)	7.18 slack (MET)	5
12	y (Input Port)	475 (Flip-Flop)	7.18 slack (MET)	5
13	y (Input Port)	487 (Flip-Flop)	7.18 slack (MET)	5
14	y (Input Port)	447 (Flip-Flop)	7.18 slack (MET)	5
15	y (Input Port)	451 (Flip-Flop)	7.18 slack (MET)	5
16	y (Input Port)	501 (Flip-Flop)	7.18 slack (MET)	5
17	y (Input Port)	443 (Flip-Flop)	7.18 slack (MET)	5
18	y (Input Port)	449 (Flip-Flop)	7.18 slack (MET)	5
19	y (Input Port)	495 (Flip-Flop)	7.18 slack (MET)	5
20	y (Input Port)	497 (Flip-Flop)	7.18 slack (MET)	5

OpenLane Container (ff5509f):/openlane/designs/spm/runs/RUN_2026.05.23_12.49.50/reports/routing\$ cat 21-grt_sta.max.rpt

```

=====
report_checks -path_delay max (Setup)
=====
===== Typical Corner =====
Startpoint: _444_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: p (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout  Cap  Slew  Delay  Time  Description
-----
          0.00  0.00  clock clk (rise edge)
          0.00  0.00  clock source latency
          0.04  0.03  0.03 ^ clk (in)
1  0.01      0.04  0.00  0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
          0.08  0.17  0.20 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
8  0.06      0.08  0.00  0.20 ^ clkbuf_3_0_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
          0.04  0.14  0.34 ^ clkbuf_3_0_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
9  0.02      0.04  0.00  0.34 ^ _444_/CLK (sky130_fd_sc_hd_dfrtp_1)
          0.04  0.36  0.70 v _444_/Q (sky130_fd_sc_hd_dfrtp_1)
1  0.00      0.04  0.00  0.70 v output35/A (sky130_fd_sc_hd_clkbuf_4)
          0.08  0.19  0.89 v output35/X (sky130_fd_sc_hd_clkbuf_4)
1  0.03      0.08  0.00  0.89 v p (out)
          0.89  0.89  data arrival time

          10.00 10.00  clock clk (rise edge)
          0.00 10.00  clock network delay (propagated)
         -0.25  9.75  clock uncertainty
          0.00  9.75  clock reconvergence pessimism
         -2.00  7.75  output external delay

```



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				7.75	data required time
				-0.89	data arrival time
				6.86	slack (MET)

Startpoint: y (input port clocked by clk)					
Endpoint: _467_ (rising edge-triggered flip-flop clocked by clk)					
Path Group: clk					
Path Type: max					
Fanout	Cap	Slew	Delay	Time	Description

			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
			0.00	2.00	v y (in)
1	0.00	0.01	0.00	2.00	y (net)
		0.01	0.00	2.00	v input34/A (sky130_fd_sc_hd__clkbuf_1)
		0.02	0.07	2.07	v input34/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net34 (net)
		0.02	0.00	2.07	v _206_/A (sky130_fd_sc_hd__clkbuf_4)
		0.06	0.16	2.23	v _206_/X (sky130_fd_sc_hd__clkbuf_4)
10	0.02				_097_ (net)
		0.06	0.00	2.23	v _207_/A (sky130_fd_sc_hd__buf_2)
		0.06	0.17	2.41	v _207_/X (sky130_fd_sc_hd__buf_2)
10	0.02				_098_ (net)
		0.06	0.00	2.41	v _264_/A (sky130_fd_sc_hd__clkbuf_4)
		0.06	0.18	2.59	v _264_/X (sky130_fd_sc_hd__clkbuf_4)
10	0.02				_133_ (net)
		0.06	0.00	2.59	v _272_/A1 (sky130_fd_sc_hd__a3io_1)
		0.03	0.18	2.77	v _272_/X (sky130_fd_sc_hd__a3io_1)
1	0.00				_002_ (net)
		0.03	0.00	2.77	v _467_/D (sky130_fd_sc_hd__dfrtp_1)
				2.77	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
		0.04	0.03	10.03	^ clk (in)
1	0.01				clk (net)
		0.04	0.00	10.03	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.08	0.15	10.18	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
8	0.06				clknet_0_clk (net)
		0.08	0.00	10.18	^ clkbuf_3_1_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.12	10.30	^ clkbuf_3_1_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
6	0.01				clknet_3_1_leaf_clk (net)
		0.03	0.00	10.30	^ _467_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	10.05	clock uncertainty
			0.00	10.05	clock reconvergence pessimism
			-0.11	9.94	library setup time
				9.94	data required time
				9.94	data required time
				-2.77	data arrival time
				7.17	slack (MET)

Startpoint: y (input port clocked by clk)					
Endpoint: _471_ (rising edge-triggered flip-flop clocked by clk)					
Path Group: clk					
Path Type: max					
Fanout	Cap	Slew	Delay	Time	Description

			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
		0.01	0.00	2.00	v y (in)
1	0.00				y (net)
		0.01	0.00	2.00	v input34/A (sky130_fd_sc_hd__clkbuf_1)
		0.02	0.07	2.07	v input34/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net34 (net)
		0.02	0.00	2.07	v _206_/A (sky130_fd_sc_hd__clkbuf_4)



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```

Startpoint: rst (input port clocked by clk)
Endpoint: _504_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout  Cap  Slew  Delay  Time  Description
-----
          0.00  0.00  0.00  0.00  clock clk (rise edge)
          0.00  0.00  0.00  0.00  clock network delay (propagated)
          2.00  2.00  v  input external delay
          0.01  0.00  2.00  v  rst (in)
1  0.00          0.01  0.00  2.00  v  rst (net)
          0.01  0.00  2.00  v  input1/A (sky130_fd_sc_hd_clkbuf_1)
          0.02  0.07  2.07  v  input1/X (sky130_fd_sc_hd_clkbuf_1)
1  0.00          0.02  0.00  2.07  v  net1 (net)
          0.02  0.00  2.07  v  _372_/A (sky130_fd_sc_hd_clkbuf_4)
          0.07  0.18  2.25  v  _372_/X (sky130_fd_sc_hd_clkbuf_4)
10  0.03          0.07  0.00  2.25  v  _199_ (net)
          0.07  0.00  2.25  v  _440_/A (sky130_fd_sc_hd_inv_2)
          0.03  0.06  2.31  ^  _440_/Y (sky130_fd_sc_hd_inv_2)
1  0.00          0.03  0.00  2.31  ^  _094_ (net)
          0.03  0.00  2.31  ^  _504_/RESET_B (sky130_fd_sc_hd_dfrtp_1)
          2.31  data arrival time

          10.00  10.00  0.00  10.00  clock clk (rise edge)
          0.00  10.00  0.00  10.00  clock source latency
          0.04  0.03  10.03  ^  clk (in)
1  0.01          0.04  0.00  10.03  ^  clk (net)
          0.04  0.00  10.03  ^  clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
          0.08  0.15  10.18  ^  clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
8  0.06          0.08  0.00  10.18  ^  clknet_0_clk (net)
          0.08  0.00  10.18  ^  clkbuf_3_5_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
          0.04  0.13  10.31  ^  clkbuf_3_5_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
9  0.02          0.04  0.00  10.31  ^  clknet_3_5_leaf_clk (net)
          0.04  0.00  10.31  ^  _504_/CLK (sky130_fd_sc_hd_dfrtp_1)
          -0.25  10.06  clock uncertainty
  
```




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ASSIGNMENT-2

The domain of the Project:
PHYSICAL DESIGN (VLSI)

NAME: GURUKIRAN A L
BATCH: G4-25-VLSI



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Implementing a 4-bit Up/Down Counter using Verilog HDL and perform the complete OpenLane RTL-to-GDSII flow. The objective is to analyze timing, area, power, utilization, routing, IR drop, and maximum operating frequency.

A 4-bit Up/Down Counter increments or decrements its count value depending on the control signal.

Verilog Code: -

```
module updown_counter (  
  
    input clk,  
  
    input rst,  
  
    input up_down,  
  
    output reg [3:0] count  
  
);  
  
always @(posedge clk or posedge rst)  
  
begin  
  
    if (rst)  
  
        count <= 4'b0000;  
  
    else if (up_down)  
  
        count <= count + 1'b1;  
  
    else  
  
        count <= count - 1'b1;  
  
end  
  
endmodule
```



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cd ~/Desktop/OpenLane/designs

mkdir updown_counter

cd updown_counter

mkdir src

```
shams@VLSI:~/Desktop/OpenLane$ make mount
cd /home/shams/Desktop/OpenLane && \
  docker run --rm -v /home/shams/Desktop/OpenLane:/openlane -v /home/shams/Desktop/
all -v /home/shams:/home/shams -v /home/shams/.ciel:/home/shams/.ciel -e PDK_ROOT=/home/s
er 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt
/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -design updown_counter
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/updown_counter/config.json'...
[INFO]: PDK Root: /home/shams/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF Files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/updown_counter/runs/RUN_2026.06.06_09.22
...
[INFO]: 0 errors found by linter
[INFO]: 0 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RL
hesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/updown_counter/runs/RUN_2026.06.06_09
fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power g
[INFO]: Floorplanned with width 35.88 and height 35.36.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/updown_counter/runs/RUN_2026.06.06_09.2
...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/fla
[STEP 7]
[INFO]: Running Global Placement (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.4
...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RL
ement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/updown_counter/runs/
acement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/updown_counter/runs/RUN_2026.06.06_09.22
log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RL
ement/11-dpl_sta.log)...
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/updown_counter/runs/RUN_2026.06.06_09.
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RL
13-cts_sta.log)...
[STEP 14]
[INFO]: Running Placement Resizer Timing Optimizations (log: designs/updown_counter/runs/
```




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```
[INFO]: Running Placement Resizer Timing Optimizations (log: designs/updown_counter/runs/s/14-resizer.log)...\n[STEP 15]\n[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/updown_counter/gs/routing/15-resizer_design.log)...\n[STEP 16]\n[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/16-rsz_design_sta.log)...\n[STEP 17]\n[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/updown_counter/gs/routing/17-resizer_timing.log)...\n[STEP 18]\n[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/18-rsz_timing_sta.log)...\n[STEP 19]\n[INFO]: Running Global Routing (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/19-global_routing.log)...\n[INFO]: Starting OpenROAD Antenna Repair Iterations...\n[STEP 20]\n[INFO]: Writing Verilog (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/roa/20-verilog.log)...\n[STEP 21]\n[INFO]: Running Single-Corner Static Timing Analysis (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/21-grt_sta.log)...\n[STEP 22]\n[INFO]: Running Fill Insertion (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/22-fill_insertion.log)...\n[STEP 23]\n[INFO]: Running Detailed Routing (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/23-detailed_routing.log)...\n[INFO]: No DRC violations after detailed routing.\n[STEP 24]\n[INFO]: Checking Wire Lengths (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/24-wire_lengths.log)...\n[INFO]: Wire lengths are within tolerance.
```

```
[STEP 25]\n[INFO]: Running SPEF Extraction at the min process corner (log: designs/updown_counter/runs/signoff/25-parasitics_extraction.min.log)...\n[STEP 26]\n[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/updown_counter/runs/signoff/26-rcx_mcsta.min.log)...\n[STEP 27]\n[INFO]: Running SPEF Extraction at the max process corner (log: designs/updown_counter/runs/signoff/27-parasitics_extraction.max.log)...\n[STEP 28]\n[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/updown_counter/runs/signoff/28-rcx_mcsta.max.log)...\n[STEP 29]\n[INFO]: Running SPEF Extraction at the nom process corner (log: designs/updown_counter/runs/signoff/29-parasitics_extraction.nom.log)...\n[STEP 30]\n[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/updown_counter/runs/signoff/30-rcx_mcsta.nom.log)...\n[STEP 31]\n[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/updown_counter/runs/signoff/31-rcx_sta.log)...\n[WARNING]: Module sky130_fd_sc_hd__tapvpwrvgn1 blackboxed during sta\n[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta\n[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta\n[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta\n[STEP 32]\n[INFO]: Creating IR Drop Report (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/32-ir_drop_report.log)...\n[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.\n[STEP 33]\n[INFO]: Running Magic to generate various views...\n[INFO]: Streaming out GDSII with Magic (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/gs/routing/33-magic.log)...\n[INFO]: Magic completed successfully.
```



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```
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40/lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with KLayout (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using KLayout (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/xor.log)...
[INFO]: No XOR differences between KLayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/write_powered_verilog.log, designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/signoff/38-write_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/signoff/39-run_lvs.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/logs/signoff/40-run_drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/updown_counter/runs/RUN_2026.06.06_09.22.40/erc_screen.log)...
[INFO]: Saving current set of views in 'designs/updown_counter/runs/RUN_2026.06.06_09.22.40/views'...
```

```
[INFO]: Saving current set of views in 'designs/updown_counter/runs/RUN_2026.06.06_09.22.40/views'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/updown_counter/runs/RUN_2026.06.06_09.22.40/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/updown_counter/runs/RUN_2026.06.06_09.22.40/metrics.rpt'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid area is 0.000000.
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvngnd_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
```

OpenLane Container (ff5509f):/openlane\$



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Figuring out Maximum Frequency

```
OpenLane Container (ff5509f):/openlane/designs/updown_counter/runs/RUN_2026_06_06_09_22_40$ ls
cmds.log      config.tcl    OPENLANE_COMMIT  PDK_SOURCES  results      tnp
config_in.tcl logs          openlane.log     reports       runtime.yaml  warnings.log
OpenLane Container (ff5509f):/openlane/designs/updown_counter/runs/RUN_2026_06_06_09_22_40$ ls reports/signoff
26-mca          31-rcx_sta.skew.rpt      41-antenna_violators_pins.txt  spice.feedback.txt
28-mca          31-rcx_sta.summary.rpt   41-antenna_violators.rpt       updown_counter.rpt
30-mca          32-irdrop-VGND.rpt       drc.klayout.xml                updown_counter.rpt.debug.gz
31-rcx_sta.checks.rpt  32-irdrop-VPWR.rpt       drc.rdb                        updown_counter.rpt.error.gz
31-rcx_sta.max.rpt    35-xor.rpt               drc.rpt
31-rcx_sta.min.rpt    35-xor.xml               drc.tcl
31-rcx_sta.power.rpt  39-updown_counter.lvs.rpt drc.tr
OpenLane Container (ff5509f):/openlane/designs/updown_counter/runs/RUN_2026_06_06_09_22_40$ grep "worst slack" reports/signoff/31-rcx_sta.summary.rpt
worst slack 6.83
worst slack 0.32
```

Using command “grep "worst slack" reports/signoff/31-rcx_sta.summary.rpt “ I had found the **worst slack**.

worst slack- 6.83

report_worst_slack -max (Setup)

worst slack 6.83

report_worst_slack -min (Hold)

worst slack 0.32

Since maximum frequency is determined by setup timing, not hold timing.

Maximum frequency depends on the longest path (critical path):

Critical Path Delay = Clock Period - Setup Slack

Clock Period = 10 ns

Setup Slack = 6.83 ns

Critical Path Delay = 10 - 6.83

= 3.17 ns

Fmax = 1 / 3.17 ns

= 315.46 MHz



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Timing Verification Table:

Clock Period	Frequency	Slack	Result
10 ns	100 MHz	+6.83 ns	Pass
6 ns	166.67 MHz	+2.83 ns	Pass
5 ns	200 MHz	+1.83 ns	Pass
4 ns	250 MHz	+0.83 ns	Pass
3 ns	333.33 MHz	-0.17 ns	Fail

Therefore, the design can operate successfully up to approximately 315 MHz, while timing violations begin to appear when the clock period is reduced below the critical path delay of 3.17 ns.

```
shams@VLSTI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40
$ grep -i uncertainty config.tcl
set ::env(SYNTH_CLOCK_UNCERTAINTY) "0.25"
```

Uncertainty: - 0.25ns

```
shams@VLSTI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40
$ grep -i derate config.tcl
set ::env(SYNTH_TIMING_DERATE) "0.05"
```

Derate Value: - 0.05%



Worst Slack Value at Each STA Stage

```
shams@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40
$ grep -R "worst slack" logs reports/signoff
logs/placement/7-global.log:[INFO GPL-0100] worst slack 6.95e-09
logs/placement/7-global.log:[INFO GPL-0100] worst slack 6.93e-09
logs/placement/7-global.log:[INFO GPL-0100] worst slack 6.93e-09
logs/placement/7-global.log:[INFO GPL-0100] worst slack 6.91e-09
logs/placement/7-global.log:[INFO GPL-0100] worst slack 6.91e-09
logs/placement/8-gpl_sta.log:worst slack 6.88
logs/placement/8-gpl_sta.log:worst slack 0.33
logs/placement/11-dpl_sta.log:worst slack 6.86
logs/placement/11-dpl_sta.log:worst slack 0.21
logs/signoff/26-rcx_mcsta.min.log:worst slack 6.01
logs/signoff/26-rcx_mcsta.min.log:worst slack 0.11
logs/signoff/31-rcx_sta.log:worst slack 6.83
logs/signoff/31-rcx_sta.log:worst slack 0.32
logs/signoff/28-rcx_mcsta.max.log:worst slack 5.98
logs/signoff/28-rcx_mcsta.max.log:worst slack 0.12
logs/signoff/30-rcx_mcsta.nom.log:worst slack 5.99
logs/signoff/30-rcx_mcsta.nom.log:worst slack 0.11
logs/synthesis/2-sta.log:worst slack 6.91
logs/synthesis/2-sta.log:worst slack 0.32
logs/routing/21-grt_sta.log:worst slack 6.89
logs/routing/21-grt_sta.log:worst slack 0.27
logs/routing/18-rsz_timing_sta.log:worst slack 6.83
logs/routing/18-rsz_timing_sta.log:worst slack 0.32
logs/routing/16-rsz_design_sta.log:worst slack 6.83
logs/routing/16-rsz_design_sta.log:worst slack 0.32
logs/cts/13-cts_sta.log:worst slack 6.83
logs/cts/13-cts_sta.log:worst slack 0.20
reports/signoff/30-mca/rcx_nom_sta.summary.rpt:worst slack 5.99
reports/signoff/30-mca/rcx_nom_sta.summary.rpt:worst slack 0.11
reports/signoff/31-rcx_sta.summary.rpt:worst slack 6.83
reports/signoff/31-rcx_sta.summary.rpt:worst slack 0.32
reports/signoff/28-mca/rcx_max_sta.summary.rpt:worst slack 5.98
reports/signoff/28-mca/rcx_max_sta.summary.rpt:worst slack 0.12
reports/signoff/26-mca/rcx_min_sta.summary.rpt:worst slack 6.01
reports/signoff/26-mca/rcx_min_sta.summary.rpt:worst slack 0.11
```

STA Stage	Setup Worst Slack (ns)	Hold Worst Slack (ns)
Synthesis STA	6.91	0.32
Global Placement STA	6.88	0.33
Detailed Placement STA	6.86	0.21
CTS STA	6.83	0.20
Routing Design Optimization STA	6.83	0.32
Routing Timing Optimization STA	6.83	0.32
Global Routing STA	6.89	0.27
Signoff STA	6.83	0.32



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Additional Table for Signoff Multi-Corner STA

Corner	Setup Slack (ns)	Hold Slack (ns)
Min Corner	6.01	0.11
Nominal Corner	5.99	0.11
Max Corner	5.98	0.12
Final Signoff STA	6.83	0.32

Power Analysis at Different Design Stages

```
shans@VLSI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40$ for f in reports/
o
echo "==== $f ====="
grep "Total" $f
done
==== reports/cts/13-cts_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          8.11e-05  2.78e-05  2.50e-10  1.09e-04 100.0%
==== reports/placement/11-dpl_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          3.03e-05  1.85e-05  2.13e-10  4.88e-05 100.0%
==== reports/placement/8-gpl_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          2.86e-05  2.00e-05  2.05e-10  4.86e-05 100.0%
==== reports/routing/16-rsz_design_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          8.42e-05  3.03e-05  2.57e-10  1.15e-04 100.0%
==== reports/routing/18-rsz_timing_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          8.42e-05  3.03e-05  2.57e-10  1.15e-04 100.0%
==== reports/routing/21-grt_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          8.41e-05  2.49e-05  2.57e-10  1.09e-04 100.0%
==== reports/signoff/31-rcx_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
Total          8.43e-05  3.06e-05  3.74e-10  1.15e-04 100.0%
==== reports/synthesis/2-syn_sta.power.rpt =====
Group          Internal  Switching  Leakage    Total
shans@VLSI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40$ grep "Total" rep
yn_sta.power.rpt
Group          Internal  Switching  Leakage    Total
Total          2.87e-05  1.87e-05  1.21e-10  4.74e-05 100.0%
```



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Stage	Internal Power (W)	Switching Power (W)	Leakage Power (W)	Total Power (W)
Synthesis STA	2.87E-05	1.87E-05	1.21E-10	4.74E-05
Global Placement STA	2.86E-05	2.00E-05	2.05E-10	4.86E-05
Detailed Placement STA	3.03E-05	1.85E-05	2.13E-10	4.88E-05
CTS STA	8.11E-05	2.78E-05	2.50E-10	1.09E-04
Routing Design Optimization STA	8.42E-05	3.03E-05	2.57E-10	1.15E-04
Routing Timing Optimization STA	8.42E-05	3.03E-05	2.57E-10	1.15E-04
Global Routing STA	8.41E-05	2.49E-05	2.57E-10	1.09E-04
Signoff STA	8.43E-05	3.06E-05	3.74E-10	1.15E-04

The power consumption of the 4-bit Up/Down Counter increases from $4.74 \times 10^{-5} \text{ W}$ at synthesis to $1.15 \times 10^{-4} \text{ W}$ at signoff. The major increase occurs after Clock Tree Synthesis due to clock buffer insertion. Leakage power remains negligible, while internal power is the dominant contributor. The final signoff analysis confirms that the design is low-power and timing-clean.

Area of Design at Each Stage

```
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.48$ grep -R "Design area" logs reports
logs/placement/8-gpl_sta.log:Design area 277 u^2 22% utilization.
logs/placement/11-dpl_sta.log:Design area 259 u^2 20% utilization.
logs/placement/9-resizer.log:Design area 259 u^2 20% utilization.
logs/routing/21-grt_sta.log:Design area 353 u^2 28% utilization.
logs/routing/15-resizer_design.log:Design area 353 u^2 28% utilization.
logs/routing/18-rsz_timing_sta.log:Design area 353 u^2 28% utilization.
logs/routing/16-rsz_design_sta.log:Design area 353 u^2 28% utilization.
logs/routing/17-resizer_timing.log:Design area 353 u^2 28% utilization.
logs/cts/13-cts_sta.log:Design area 334 u^2 26% utilization.
logs/cts/14-resizer.log:Design area 354 u^2 28% utilization.
```




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Stage	Design Area (μm^2)
Synthesis	257.75
Global Placement	277
Detailed Placement	259
CTS	334
CTS Optimization	354
Routing Design Optimization	353
Routing Timing Optimization	353
Global Routing	353
Final Signoff	353

The area of the Up/Down Counter design increases gradually throughout the OpenLane flow due to optimization and clock tree insertion. The synthesized design starts with an area of **257.75 μm^2** . During placement and CTS, additional cells such as buffers are introduced, causing the area to increase. After routing and optimization, the design area stabilizes at **353 μm^2** . The final area indicates that the design successfully meets physical design requirements while maintaining efficient utilization and sufficient routing space.

Signal Route Length

```
shans@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2020-05-06_09:22:41$ grep -R "wire length" logs/routing
logs/routing/23-detailed.log:Total wire length = 482 um.
logs/routing/23-detailed.log:Total wire length on LAYER li1 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net1 = 250 um.
logs/routing/23-detailed.log:Total wire length on LAYER net2 = 217 um.
logs/routing/23-detailed.log:Total wire length on LAYER net3 = 14 um.
logs/routing/23-detailed.log:Total wire length on LAYER net4 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net5 = 0 um.
logs/routing/23-detailed.log:Total wire length = 483 um.
logs/routing/23-detailed.log:Total wire length on LAYER li1 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net1 = 254 um.
logs/routing/23-detailed.log:Total wire length on LAYER net2 = 214 um.
logs/routing/23-detailed.log:Total wire length on LAYER net3 = 14 um.
logs/routing/23-detailed.log:Total wire length on LAYER net4 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net5 = 0 um.
logs/routing/23-detailed.log:Total wire length = 483 um.
logs/routing/23-detailed.log:Total wire length on LAYER li1 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net1 = 254 um.
logs/routing/23-detailed.log:Total wire length on LAYER net2 = 214 um.
logs/routing/23-detailed.log:Total wire length on LAYER net3 = 14 um.
logs/routing/23-detailed.log:Total wire length on LAYER net4 = 0 um.
logs/routing/23-detailed.log:Total wire length on LAYER net5 = 0 um.
logs/routing/15-resizer.design.log:[INFO RSZ-0058] Using max wire length 4508um.
```

Total route length = 483 μm

The total routed wire length of the design is **483 μm** , which is much smaller than the resizer threshold of **4508 μm** . Therefore, no long-wire optimization was required, indicating that the design is compact and efficiently routed.



```
shams@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22
.40$ grep -i clk reports/routing/24-wire_lengths.csv
clknet_1_1__leaf_clk,27.72
clknet_1_0__leaf_clk,22.88
clk,22.03
clknet_0_clk,15.68
```

Total Clock Route Length

$$27.72 + 22.88 + 22.03 + 15.68 = 88.31 \mu m$$

Total Signal Route Length

$$483 - 88.31 = 394.69 \mu m$$

OpenLane generates a file named 24-wire_lengths.csv containing the routed length of each net. All nets containing the keyword clk were extracted and summed to obtain the total clock route length. The signal route length was then calculated by subtracting the clock route length from the total wire length reported by OpenLane.

The total routed wire length of the design is **483 μm** . Out of this, **88.31 μm** is used by clock distribution networks and **394.69 μm** is used by signal routing.

Design Utilization at Different Stages

```
shams@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22
.40$ grep -R "Design area" logs
logs/placement/8-gpl_sta.log:Design area 277 u^2 22% utilization.
logs/placement/11-dpl_sta.log:Design area 259 u^2 20% utilization.
logs/placement/9-resizer.log:Design area 259 u^2 20% utilization.
logs/routing/21-grt_sta.log:Design area 353 u^2 28% utilization.
logs/routing/15-resizer_design.log:Design area 353 u^2 28% utilization.
logs/routing/18-rsz_timing_sta.log:Design area 353 u^2 28% utilization.
logs/routing/16-rsz_design_sta.log:Design area 353 u^2 28% utilization.
logs/routing/17-resizer_timing.log:Design area 353 u^2 28% utilization.
logs/cts/13-cts_sta.log:Design area 334 u^2 26% utilization.
logs/cts/14-resizer.log:Design area 354 u^2 28% utilization.
```



Stage	Utilization (%)
Global Placement (GPL)	22
Detailed Placement (DPL)	20
Clock Tree Synthesis (CTS)	26
Post-CTS Optimization	28
Routing	28
Signoff (Final)	28

Floorplan Analysis

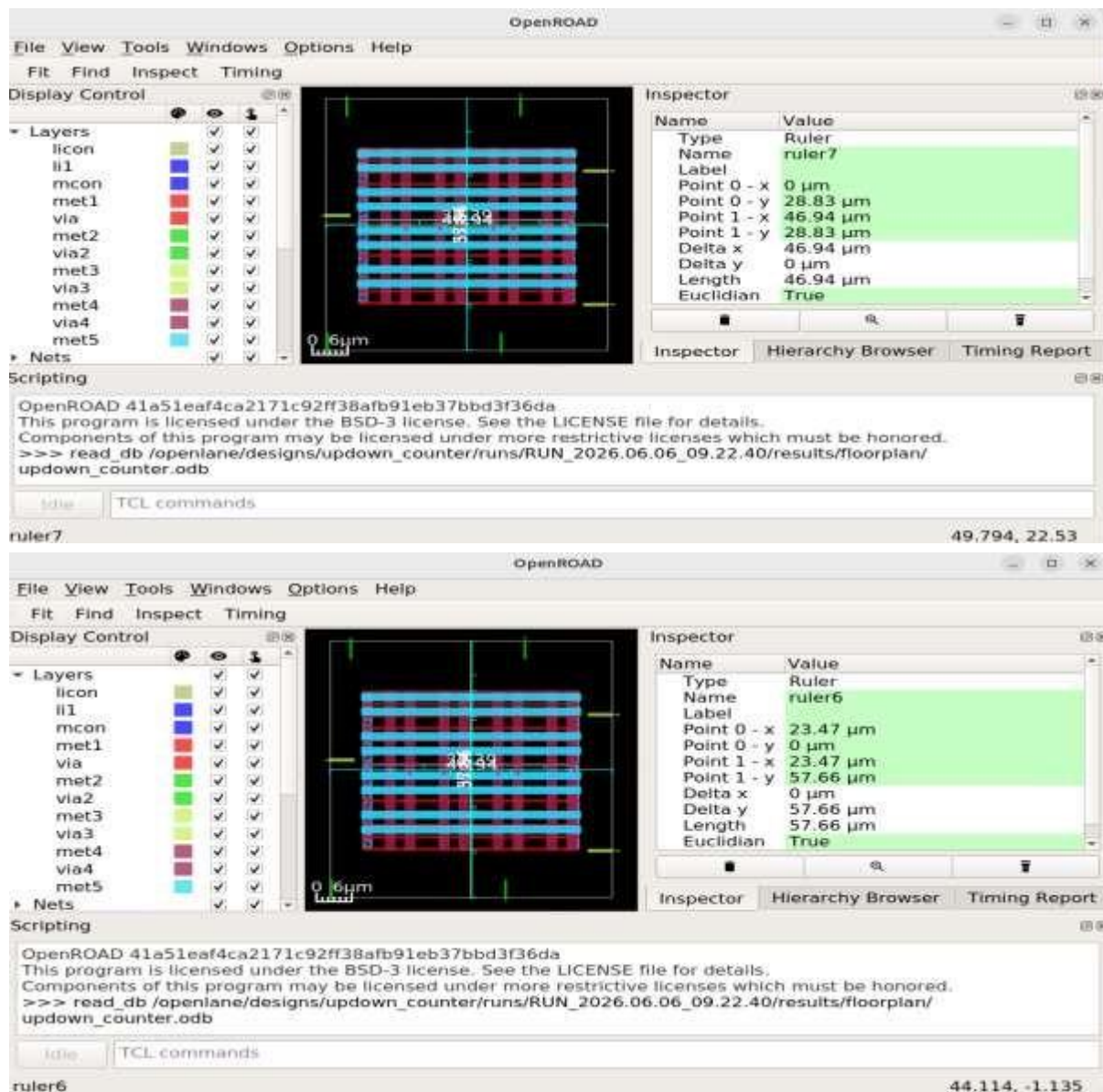
Floorplan Layout

The top screenshot shows the OpenROAD floorplan layout tool. The main window displays a floorplan with a ruler labeled 'ruler3' and a length of 35.39 μm. The Layers panel on the left shows various layers like licon, li1, mcon, met1, via, met2, via2, met3, via3, met4, via4, and met5. The Scripting panel at the bottom shows the command 'read_db /openlane/designs/updown_counter/runs/RUN_2026.06.06_09.22.40/results/floorplan/updown_counter.odb'. The Inspector panel on the right shows the ruler's properties, including its name, type, and length.

The bottom screenshot shows the same OpenROAD floorplan layout tool, but with a different ruler labeled 'ruler4' and a length of 35.84 μm. The Layers panel and Scripting panel are identical to the top screenshot. The Inspector panel on the right shows the ruler's properties, including its name, type, and length.



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Description: The floorplan of the up-down counter design was analyzed using the OpenROAD GUI. The floorplan stage defines the die area, core area, placement rows, power distribution network, and physical-only cells required for fabrication. The layout was inspected using the GUI, and measurements of the core and die dimensions were obtained using the Ruler tool.

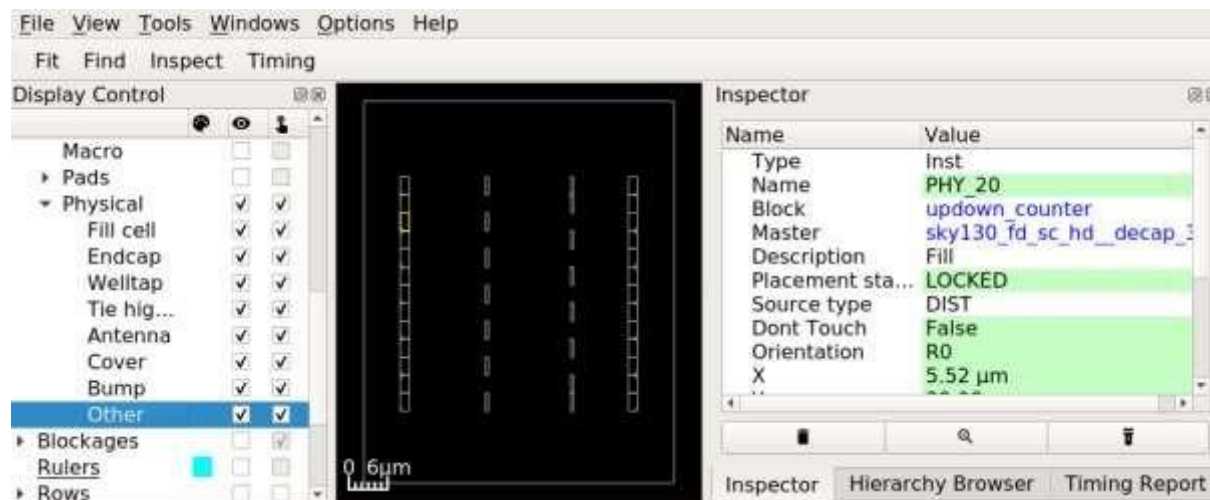
Parameter	Value (μm)
Core Width	35.39
Core Height	35.84
Die Width	46.94
Die Height	57.66



The measured core dimensions are $35.39\ \mu\text{m} \times 35.84\ \mu\text{m}$, providing the placement region for standard cells. The overall die dimensions are $46.94\ \mu\text{m} \times 57.66\ \mu\text{m}$, which include the core area, power network structures, and boundary spacing required for fabrication. The die area is larger than the core area to accommodate routing resources and physical-only cells.

Physical-Only Cells at Floorplan Stage

Physical-only cells were visualized using the OpenROAD GUI by enabling the Physical cell display option. These cells do not perform logical operations but are inserted to improve manufacturability, reliability, and power integrity.



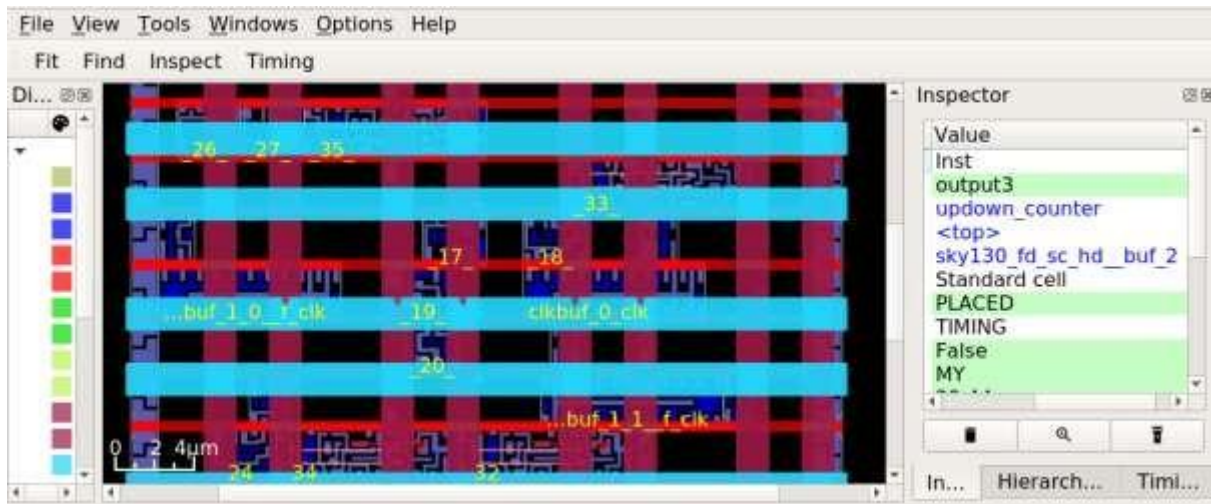
Physical Cell Type	Purpose
Fill Cell	Fills gaps between standard cells and maintains well continuity
Endcap Cell	Protects row boundaries and satisfies design rules
Well Tap Cell	Prevents latch-up by connecting wells to power rails
Decap Cell	Improves power integrity and reduces voltage fluctuations
Antenna Cell	Prevents antenna violations during fabrication

The OpenROAD GUI shows the presence of fill cells, endcap cells, well tap cells, and decap cells. These cells are inserted automatically during physical implementation to ensure reliable chip fabrication and robust power distribution.



Placement Analysis

Buffer: -

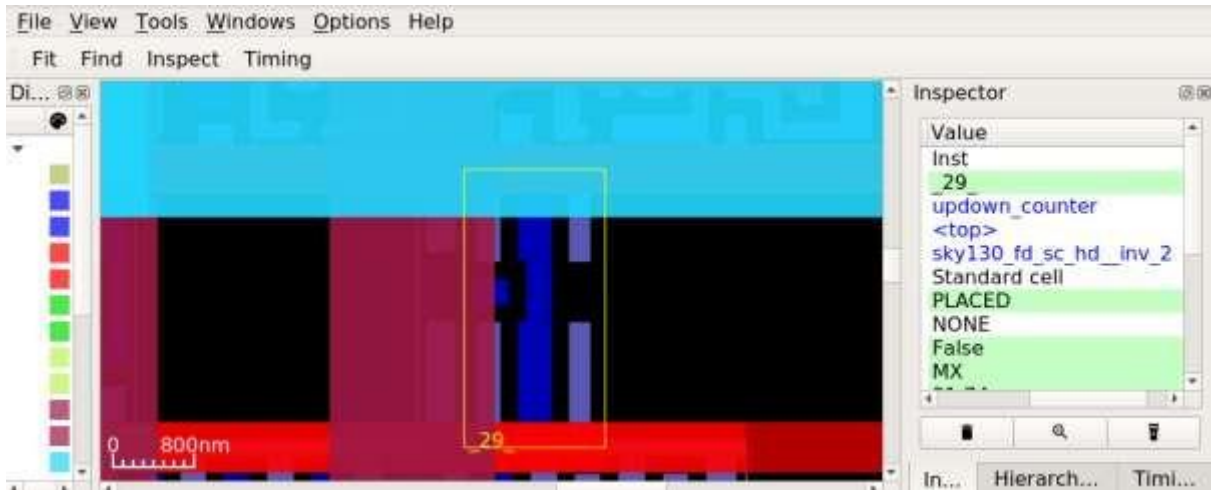


Buffer Cells Identified

Instance Name	Cell Type
hold1	sky130_fd_sc_hd__buf_1
clkbuf_0_clk	sky130_fd_sc_hd__clkbuf_16
clkbuf_1_0__f_clk	sky130_fd_sc_hd__clkbuf_16
clkbuf_1_1__f_clk	sky130_fd_sc_hd__clkbuf_16



Inverter: -



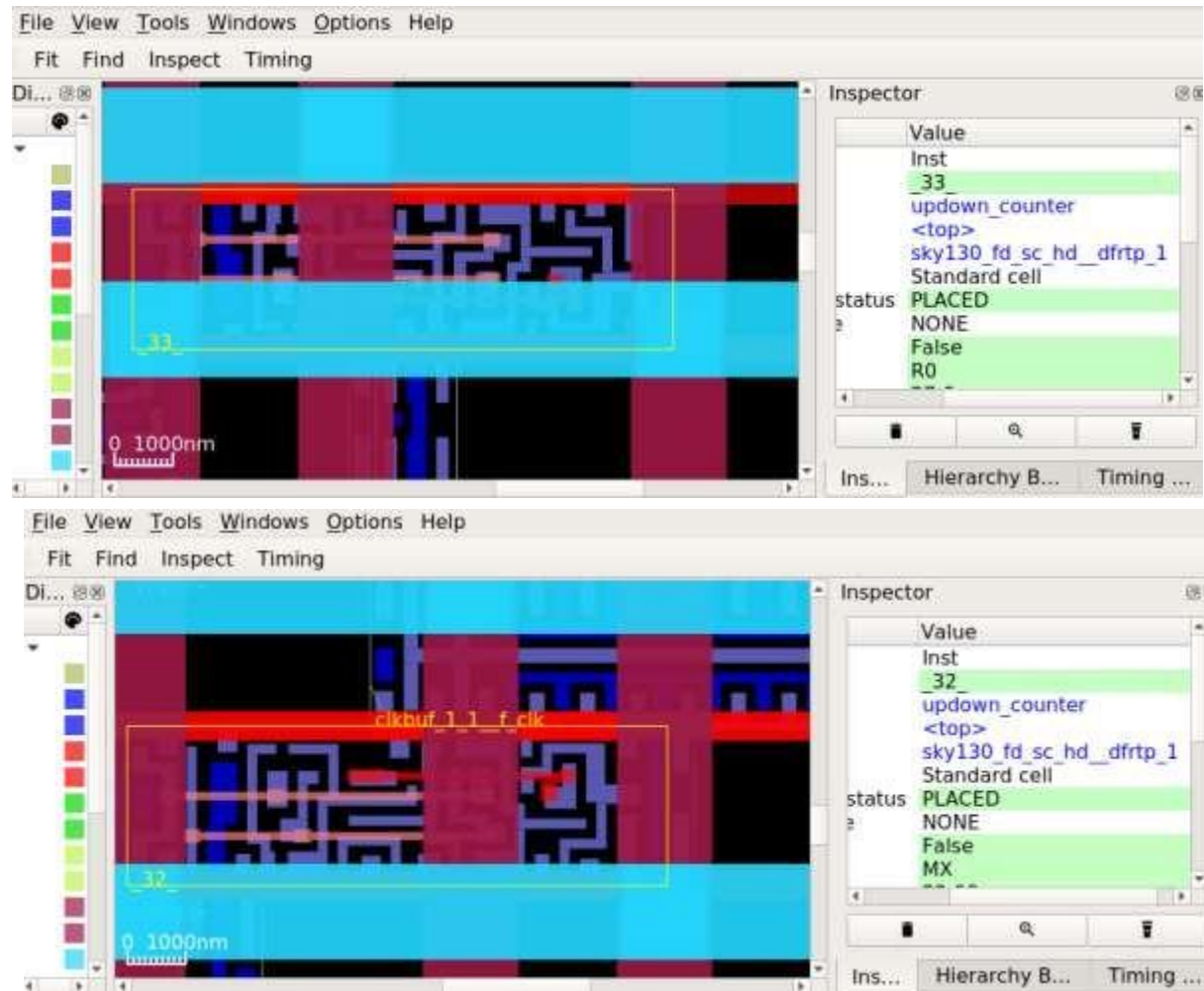
Inverter Cells Identified

Instance Name	Cell Type
16	sky130_fd_sc_hd __nv_2
28	sky130_fd_sc_hd __nv_2
29	sky130_fd_sc_hd __nv_2
30	sky130_fd_sc_hd __nv_2
31	sky130_fd_sc_hd __nv_2

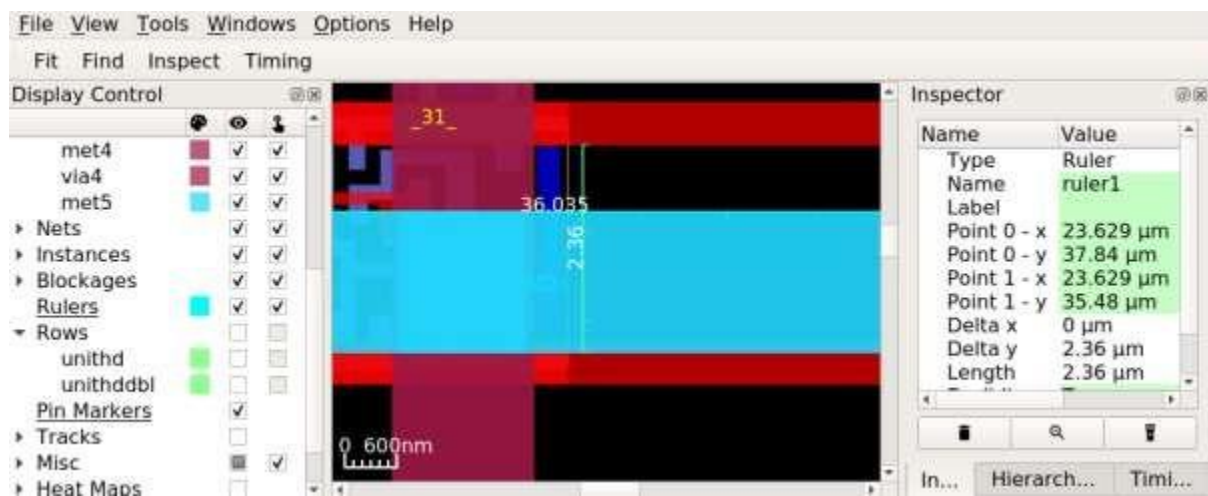


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Flops: -



Site Row: -





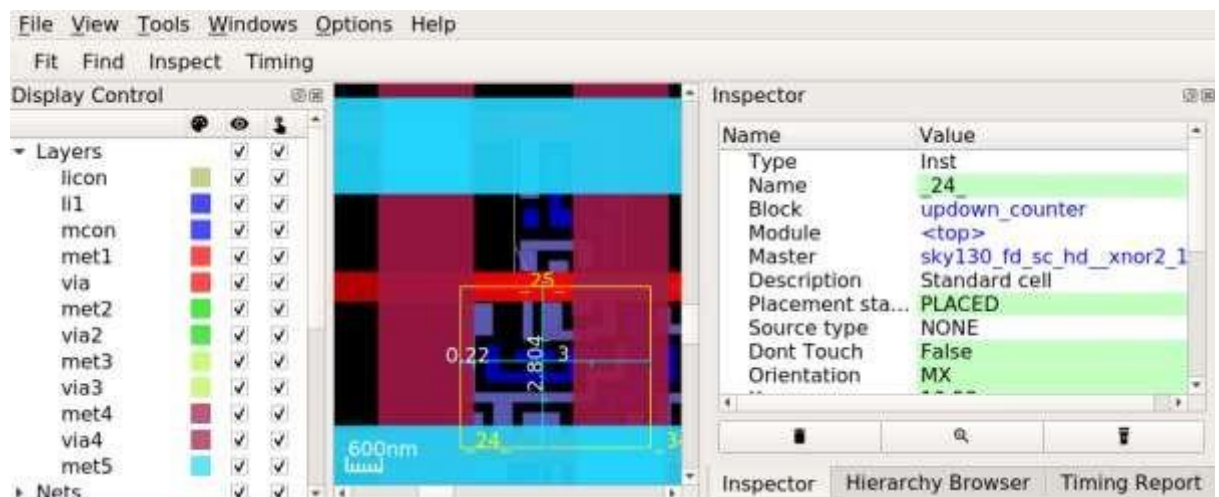
Site Row Measurement

The site row dimensions were measured using the OpenROAD GUI ruler tool during the placement stage.

Measurement	Value (μm)
Site Row Width	36.035
Site Row Height	2.36

Different library cell Height and Width

1. Xnor

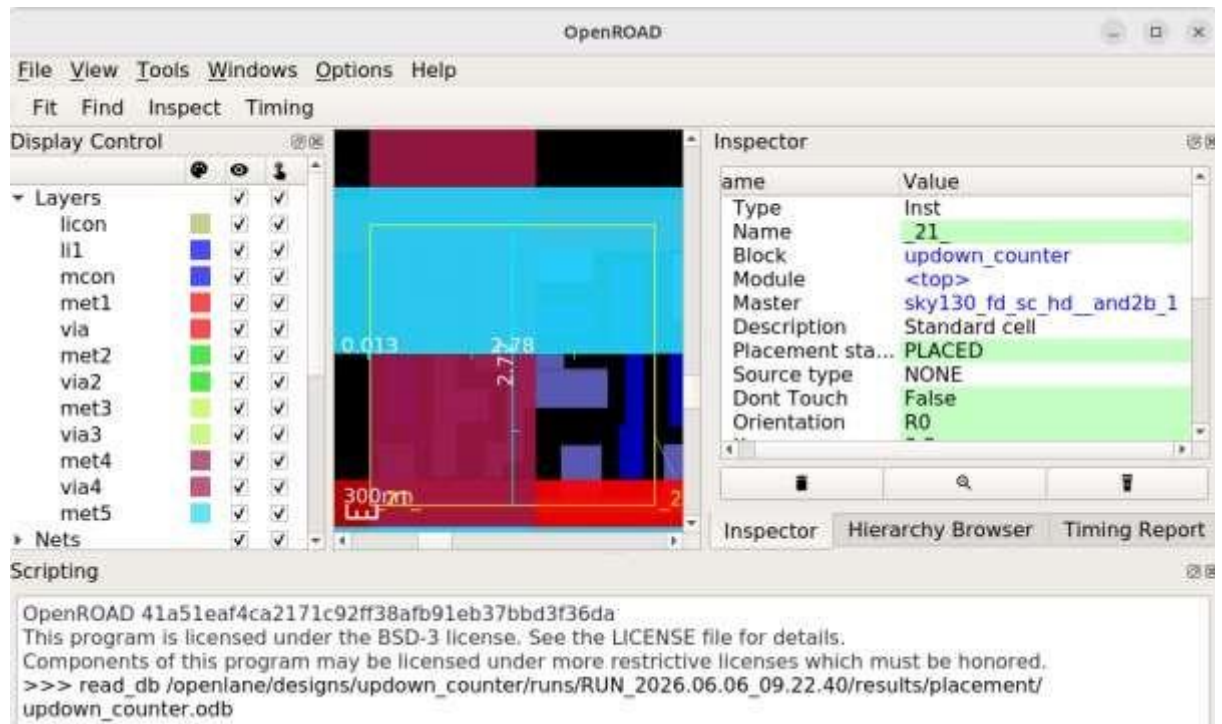


Height = 2.804 μm

Width = 3.022 μm



2. And



Height = 2.78 μm

Width = 2.72 μm

CTS Analysis: -

Synthesis Netlist

Contains only:

sky130_fd_sc_hd_inv_2_16_
sky130_fd_sc_hd_inv_2_28_
sky130_fd_sc_hd_inv_2_29_
sky130_fd_sc_hd_inv_2_30_
sky130_fd_sc_hd_inv_2_31_

```
shams@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.4
0$ grep -i "buf\|inv" results/synthesis/*.v
sky130_fd_sc_hd_inv_2_16_ (
sky130_fd_sc_hd_inv_2_28_ (
sky130_fd_sc_hd_inv_2_29_ (
sky130_fd_sc_hd_inv_2_30_ (
sky130_fd_sc_hd_inv_2_31_ (
```



Final Netlist

Contains:

sky130_fd_sc_hd__inv_2 _16_
sky130_fd_sc_hd__inv_2 _28_
sky130_fd_sc_hd__inv_2 _29_
sky130_fd_sc_hd__inv_2 _30_
sky130_fd_sc_hd__inv_2 _31_

sky130_fd_sc_hd__buf_1 hold1

sky130_fd_sc_hd__clkbuf_16 clkbuf_0_clk
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_0_f_clk
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_1__f_clk

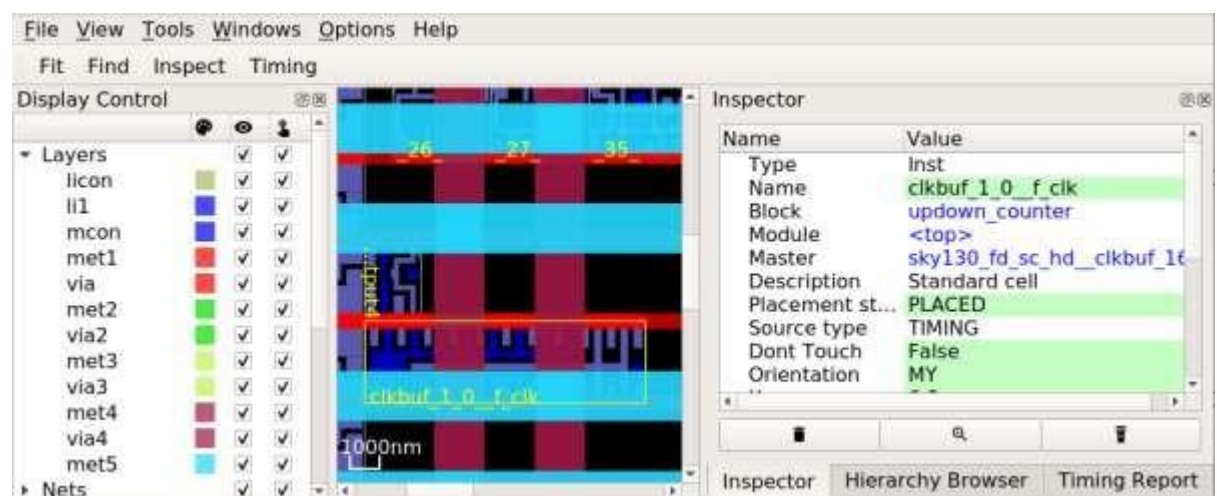
sky130_fd_sc_hd__clkbuf_2 input1
sky130_fd_sc_hd__clkbuf_2 input2

sky130_fd_sc_hd__clkbuf_4 output3
sky130_fd_sc_hd__clkbuf_4 output4
sky130_fd_sc_hd__clkbuf_4 output5
sky130_fd_sc_hd__clkbuf_4 output6

```
shams@VLSI: ~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.4
0$ grep -i "buf\\|inv" results/final/verilog/gl/updown_counter.v
sky130_fd_sc_hd__inv_2 _16_ (.A(net7),
sky130_fd_sc_hd__inv_2 _28_ (.A(net1),
sky130_fd_sc_hd__inv_2 _29_ (.A(net1),
sky130_fd_sc_hd__inv_2 _30_ (.A(net1),
sky130_fd_sc_hd__inv_2 _31_ (.A(net1),
sky130_fd_sc_hd__clkbuf_16 clkbuf_0_clk (.A(clk),
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_0_f_clk (.A(clknet_0_clk),
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_1__f_clk (.A(clknet_0_clk),
sky130_fd_sc_hd__buf_1 hold1 (.A(net3),
sky130_fd_sc_hd__clkbuf_2 input1 (.A(rst),
sky130_fd_sc_hd__clkbuf_2 input2 (.A(up_down),
sky130_fd_sc_hd__clkbuf_4 output3 (.A(net3),
sky130_fd_sc_hd__clkbuf_4 output4 (.A(net4),
sky130_fd_sc_hd__clkbuf_4 output5 (.A(net5),
sky130_fd_sc_hd__clkbuf_4 output6 (.A(net6),
```

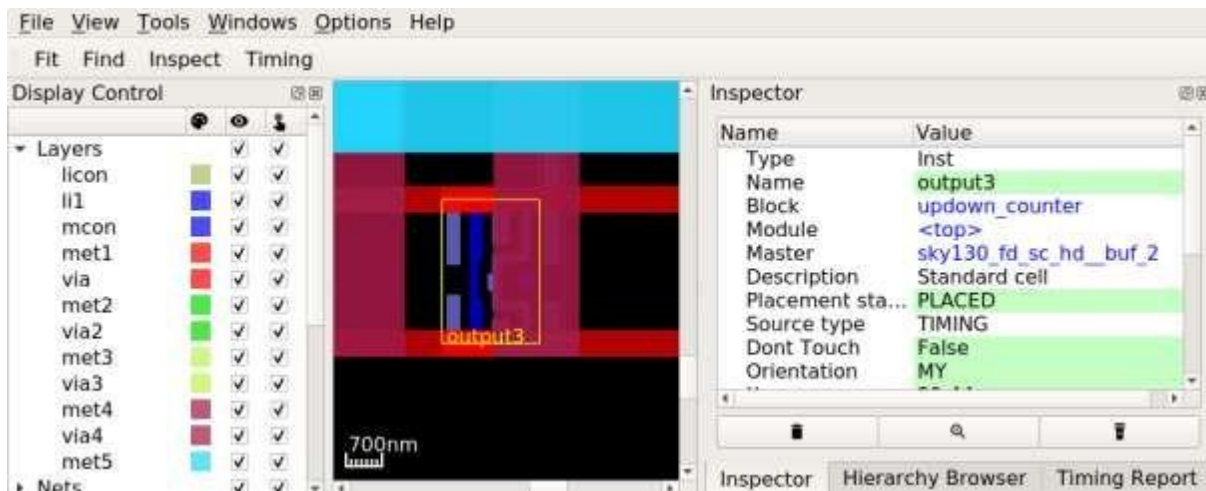


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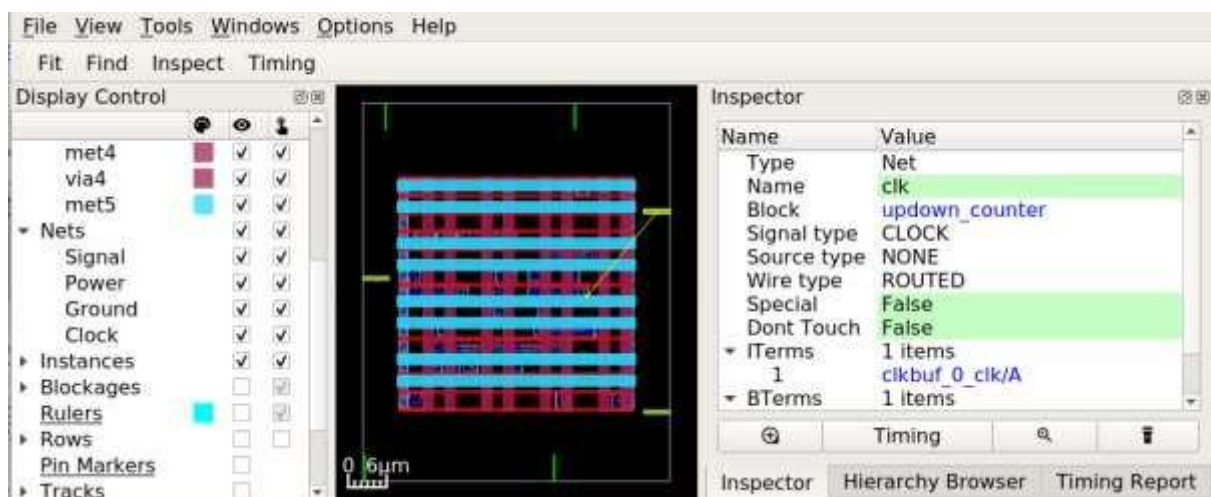




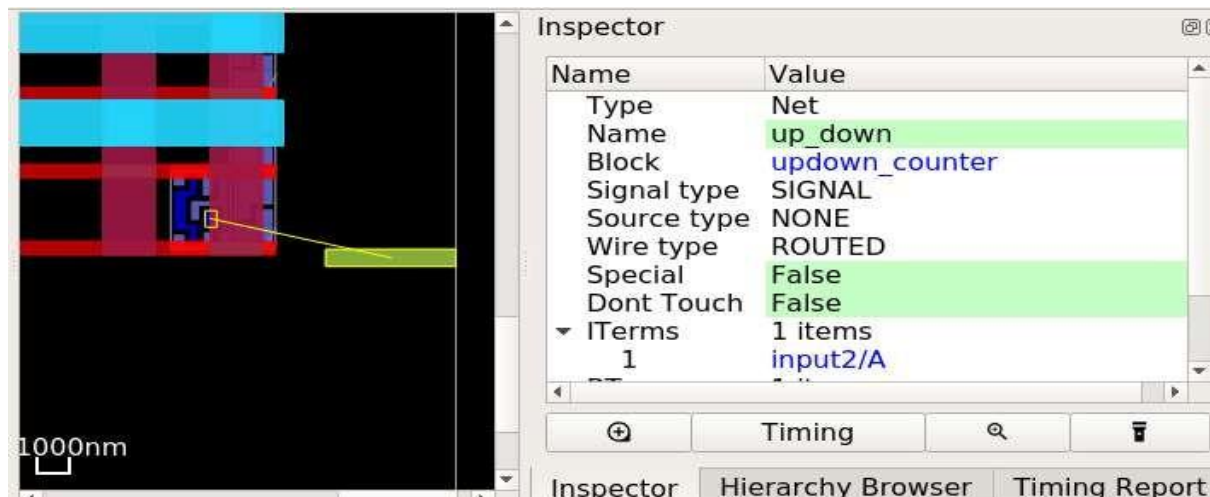
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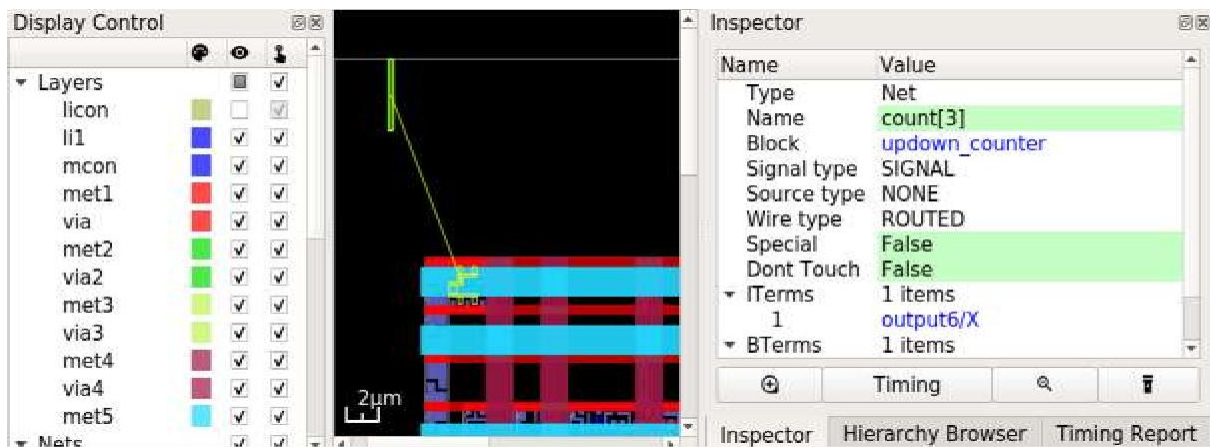


Clock Routing and its metal layer: -



Signal Routing and its metal layer: -





Newly added buffer/inverter during routing stage: -

Earlier,

```
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.4
0$ grep -i "buf\|inv" results/synthesis/*.v
sky130_fd_sc_hd__inv_2_16_ (
sky130_fd_sc_hd__inv_2_28_ (
sky130_fd_sc_hd__inv_2_29_ (
sky130_fd_sc_hd__inv_2_30_ (
sky130_fd_sc_hd__inv_2_31_ (
```

Only 5 inverters are present during synthesis stage.

After synthesis,

```
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter/runs/RUN_2026.06.06_09.22.4
0$ grep -i "buf\|inv" results/final/verilog/gl/updown_counter.v
sky130_fd_sc_hd__inv_2_16_ (.A(net7),
sky130_fd_sc_hd__inv_2_28_ (.A(net1),
sky130_fd_sc_hd__inv_2_29_ (.A(net1),
sky130_fd_sc_hd__inv_2_30_ (.A(net1),
sky130_fd_sc_hd__inv_2_31_ (.A(net1),
sky130_fd_sc_hd__clkbuf_16 clkbuf_0_clk (.A(clk),
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_0__f_clk (.A(clknet_0_clk),
sky130_fd_sc_hd__clkbuf_16 clkbuf_1_1__f_clk (.A(clknet_0_clk),
sky130_fd_sc_hd__buf_1 hold1 (.A(net3),
sky130_fd_sc_hd__clkbuf_2 input1 (.A(rst),
sky130_fd_sc_hd__clkbuf_2 input2 (.A(up_down),
sky130_fd_sc_hd__clkbuf_4 output3 (.A(net3),
sky130_fd_sc_hd__clkbuf_4 output4 (.A(net4),
sky130_fd_sc_hd__clkbuf_4 output5 (.A(net5),
sky130_fd_sc_hd__clkbuf_4 output6 (.A(net6),
```

Routing Stage inserted 7 new buffers and 0 inverters.



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32-bit Up-Down Counter

```
shams@VLSI:~/Desktop$ cd OpenLane/designs
shams@VLSI:~/Desktop/OpenLane/designs$ mkdir updown_counter_32
shams@VLSI:~/Desktop/OpenLane/designs$ cd updown_counter_32
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ mkdir src
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ ls
src
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ nano src/updown_counter_32.v
```

```
GNU nano 7.2 src/updown_counter_32.v *
module updown_counter_32 (
    input clk,
    input rst,
    input up_down,
    output reg [31:0] count
);

always @(posedge clk or posedge rst)
begin
    if (rst)
        count <= 32'd0;
    else if (up_down)
        count <= count + 1'b1;
    else
        count <= count - 1'b1;
end

endmodule

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
shams@VLSI:~/Desktop$ cd OpenLane/designs
shams@VLSI:~/Desktop/OpenLane/designs$ mkdir updown_counter_32
shams@VLSI:~/Desktop/OpenLane/designs$ cd updown_counter_32
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ mkdir src
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ ls
src
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ nano src/updown_counter_32.v
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ nano src/updown_counter_32.v
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ nano config.json
```



```
GNU nano 7.2                                config.json
"DESIGN_NAME": "updown_counter_32",
"VERILOG_FILES": "dir::src/updown_counter_32.v",
"CLOCK_PORT": "clk",
"CLOCK_PERIOD": 6,
"FP_CORE_UTIL": 70,
"FP_PDN_CORE_RING": true,
"CTS_TARGET_SKEW": 0.3,
"FP_TAPCELL_DIST": 14

[ Read 10 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

shams@VLSTI:~/Desktop/OpenLane/designs/updown_counter_32$ nano config.json
shams@VLSTI:~/Desktop/OpenLane/designs/updown_counter_32$ nano config.json
shams@VLSTI:~/Desktop/OpenLane/designs/updown_counter_32$ nano src/base.sdc
```

```
GNU nano 7.2                                src/base.sdc *
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.6 [get_clocks clk]

set_timing_derate -early 0.97
set_timing_derate -late 1.03

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```




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```
shams@VLSI:~/Desktop/OpenLane$ make mount
cd /home/shams/Desktop/OpenLane && \
    docker run --rm -v /home/shams/Desktop/OpenLane:/openlane -v /home/shams/Desktop/OpenLane/designs:/openlane/install -v /home/shams:/home/shams -v /home/shams/.ciel:/home/shams/.ciel -e PDK_ROOT=/home/shams/.ciel -e PDK=sky130A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -design updown_counter_32
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/updown_counter_32/config.json'...
[INFO]: PDK Root: /home/shams/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/updown_counter_32/runs/RUN_2026.06.07_07.23.41
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
```

```
[INFO]: Saving current set of views in 'designs/updown_counter_32/runs/RUN_2026.06.07_07.23.41/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/updown_counter_32/runs/RUN_2026.06.07_07.23.41/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/updown_counter_32/runs/RUN_2026.06.07_07.23.41/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd__tapvpwrvgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
```

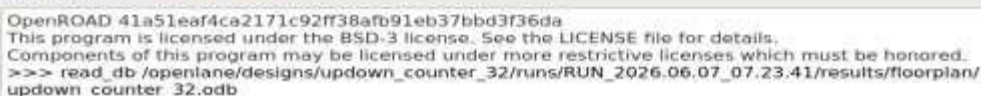
```
OpenLane Container (ff5509f):/openlane$
```


[illegible]



4. Derate

5. Insertion of two column of tap cells





6. Core Ring



7. CTS target skew 5% of the clock Period

```
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$ grep CTS_TARGET_SKEW config.json
  "CTS_TARGET_SKEW": 0.3,
shams@VLSI:~/Desktop/OpenLane/designs/updown_counter_32$
```

Clock period = 6 ns

5% of 6 ns:

$$6 \times 0.05 = 0.3 \text{ ns}$$



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ASSIGNMENT-3

The domain of the Project:
PHYSICAL DESIGN (VLSI)

NAME: GURU KIRAN AL
BATCH: G4-25-VLSI



ASSIGNMENT 3

Take Riscv 32 bit RTL

1. Uncertainty at synthesis 15% of clock period and from CTS 5% of clk period

The config.json file used is:

```
"DESIGN_NAME": "ibex_core",  
"VERILOG_FILES": "dir::rtl/ibex_top.v",  
"CLOCK_PORT": "core_clk",  
"CLOCK_PERIOD": 10,  
"SYNTH_CLOCK_UNCERTAINTY": 1.5,  
"FP_CORE_UTIL": 50,  
"PL_TARGET_DENSITY": 0.45,  
"FP_ASPECT_RATIO": 1.2,  
"FP_IO_MODE": 2,  
"SYNTH_STRATEGY": "DELAY 3",  
"RUN_CVC": 1
```

The script used to set the uncertainty is:

```
set clk_period 10.0  
set synth_uncertainty [expr {$clk_period * 0.15}]  
set cts_uncertainty [expr {$clk_period * 0.05}]  
puts "Synthesis Uncertainty: $synth_uncertainty ns"  
puts "CTS Uncertainty: $cts_uncertainty ns"
```



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After setting the clock uncertainty use the commands:

- `prep -designs riscv`
- `run_synthesis`
- `run_floorplan`
- `run_cts`
- After using `run_cts` check the CTS uncertainty is 5% of clk period if its 5% of clock period then use:
- `run_routing`

Output:

- Synthesis Uncertainty: **1.5000 ns**
- CTS Uncertainty: **0.5000 ns**
- Timing Derate (Early): **0.9500**
- Timing Derate (Late): **1.0500**



2. Set derate 3% throughout the design

The config.json must be modified to:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

After saving the config.json file use run_synthesis and then execute the command:

```
puts "\n===== DERATE CONFIGURATION ====="
exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
```

You will get the output:

```
% prep -design riscv -tag DERATE3 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/DERATE3
[INFO]: Removing existing /openlane/designs/riscv/runs/DERATE3...
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/DERATE3/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/DERATE3/logs/synthesis/2-sta.log)...
% puts $::env(RUN_DIR)
/openlane/designs/riscv/runs/DERATE3
% exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
set ::env(SYNTH_TIMING_DERATE) "0.03"
% exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
set ::env(SYNTH_TIMING_DERATE) "0.03"
% exec find $::env(RUN_DIR) -name "*.sdc"
/openlane/designs/riscv/runs/DERATE3/tmp/synthesis/synthesis.sdc
% exec grep "timing_derate" $::env(RUN_DIR)/tmp/synthesis/synthesis.sdc
child process exited abnormally
% exec cat $::env(RUN_DIR)/tmp/synthesis/synthesis.sdc
set_driving_cell sky130_fd_sc_hd__inv_2
set_load 33.442
% vi derate.tcl
% source derate.tcl

===== DERATE CONFIGURATION =====
set ::env(SYNTH_TIMING_DERATE) "0.03"
%
```




3. Set input and output delay 50% of clk time period

The updated config.json file is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The base.sdc file is created and its content is:

```
create_clock -name clk -period 6 [get_ports clk]
set_clock_uncertainty 0.9 [get_clocks clk]
set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]
~
~
```

The use run_synthesis and verify the logs and finally the output is:

```
OpenLane Container (ff5509f):/openlane$ vi designs/riscv/config.json
OpenLane Container (ff5509f):/openlane$ vi /openlane/designs/riscv/base.sdc
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design riscv -tag I050_SDC -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/I050_SDC
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/I050_SDC/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/I050_SDC/logs/synthesis/2-sta.log)...
```



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```
% exec grep -R "BASE_SDC_FILE" $::env(RUN_DIR)
/openlane/designs/riscv/runs/I050_SDC/config.tcl:set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
/openlane/designs/riscv/runs/I050_SDC/config_in.tcl:set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
% exec cat /openlane/designs/riscv/base.sdc
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.9 [get_clocks clk]

set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]
```

```
% vi logs.tcl
% source logs.tcl

===== CLOCK / IO CONSTRAINTS =====
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.9 [get_clocks clk]

set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]

===== DERATE =====
set ::env(SYNTH_TIMING_DERATE) "0.03"

===== BASE SDC =====
set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
```

From these logs we can say:

- Clock Period: 6 ns
- Clock Uncertainty: 0.9 ns
- Input Delay: 3.0 ns
- Output Delay: 3.0 ns
- Timing Derate: 0.03 (3%)

Save Point:

- Design: PicoRV32 (32-bit RISC-V)
- Clock Period: 6 ns
- Clock Uncertainty: 0.9 ns
- Input Delay: 3.0 ns
- Output Delay: 3.0 ns
- Timing Derate: 3%
- Constraint File: base.sdc



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4. Utilization should be set 60% at floorplan

The updated config.json script is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The output we get is:

```
OpenLane Container (ff5509f):/openlane$ vi designs/riscv/config.json
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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% prep -design riscv -tag UTIL60 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/UTIL60
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/UTIL60/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/UTIL60/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv/runs/UTIL60/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 408.94 and height 408.0.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv/runs/UTIL60/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv/runs/UTIL60/logs/floorplan/6-pdn.log)...
% puts [exec grep "FP_CORE_UTIL" $::env(RUN_DIR)/config.tcl]
set ::env(FP_CORE_UTIL) "60"
%
```



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Save Point:

- Design: PicoRV32 (32-bit RISC-V)
- Floorplan Utilization: 60%
- Evidence File: runs/UTIL60/config.tcl

5. Floorplan ratio should be 0.7

The updated config.json:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 0.7,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The TCL script used is:

```
puts "\n==== FLOORPLAN ASPECT RATIO ====="
puts [exec grep "FP_ASPECT_RATIO" $::env(RUN_DIR)/config.tcl]
```




The output is:

```
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% prep -design riscv -tag AR07 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/AR07
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/AR07/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/AR07/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv/runs/AR07/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 488.52 and height 340.0.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv/runs/AR07/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv/runs/AR07/logs/floorplan/6-pdn.log)...
% vi floorplan.tcl
% source floorplan.tcl

===== FLOORPLAN ASPECT RATIO =====
set ::env(FP_ASPECT_RATIO) "0.7"
%
```

6. port placement - inputs to one side outputs on other side and place it on metal4

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```



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The file used to do manual port placement is pin_order.cfg and its content is:

```
#BUS_SORT
```

```
#N
```

```
@min_distance=0.1
```

```
clk
```

```
resetn
```

```
mem_ready
```

```
pcpi_wr
```

```
pcpi_wait
```

```
pcpi_ready
```

```
irq\[0\]
```

```
irq\[1\]
```

```
irq\[2\]
```

```
irq\[3\]
```

```
irq\[4\]
```

```
irq\[5\]
```

```
irq\[6\]
```

```
irq\[7\]
```

```
irq\[8\]
```

```
irq\[9\]
```

```
irq\[10\]
```

```
irq\[11\]
```

```
irq\[12\]
```

```
irq\[13\]
```

```
irq\[14\]
```

```
irq\[15\]
```

```
irq\[16\]
```

```
irq\[17\]
```

```
irq\[18\]
```

```
irq\[19\]
```

```
irq\[20\]
```



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irq\[21\]

irq\[22\]

irq\[23\]

irq\[24\]

irq\[25\]

irq\[26\]

irq\[27\]

irq\[28\]

irq\[29\]

irq\[30\]

irq\[31\]

mem_rdata\[0\]

mem_rdata\[1\]

mem_rdata\[2\]

mem_rdata\[3\]

mem_rdata\[4\]

mem_rdata\[5\]

mem_rdata\[6\]

mem_rdata\[7\]

mem_rdata\[8\]

mem_rdata\[9\]

mem_rdata\[10\]

mem_rdata\[11\]

mem_rdata\[12\]

mem_rdata\[13\]

mem_rdata\[14\]

mem_rdata\[15\]

mem_rdata\[16\]

mem_rdata\[17\]

mem_rdata\[18\]

mem_rdata\[19\]



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mem_rdata\[20\]

mem_rdata\[21\]

mem_rdata\[22\]

mem_rdata\[23\]

mem_rdata\[24\]

mem_rdata\[25\]

mem_rdata\[26\]

mem_rdata\[27\]

mem_rdata\[28\]

mem_rdata\[29\]

mem_rdata\[30\]

mem_rdata\[31\]

pcpi_rd\[0\]

pcpi_rd\[1\]

pcpi_rd\[2\]

pcpi_rd\[3\]

pcpi_rd\[4\]

pcpi_rd\[5\]

pcpi_rd\[6\]

pcpi_rd\[7\]

pcpi_rd\[8\]

pcpi_rd\[9\]

pcpi_rd\[10\]

pcpi_rd\[11\]

pcpi_rd\[12\]

pcpi_rd\[13\]

pcpi_rd\[14\]

pcpi_rd\[15\]

pcpi_rd\[16\]

pcpi_rd\[17\]

pcpi_rd\[18\]



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pcpi_rd\[19\]

pcpi_rd\[20\]

pcpi_rd\[21\]

pcpi_rd\[22\]

pcpi_rd\[23\]

pcpi_rd\[24\]

pcpi_rd\[25\]

pcpi_rd\[26\]

pcpi_rd\[27\]

pcpi_rd\[28\]

pcpi_rd\[29\]

pcpi_rd\[30\]

pcpi_rd\[31\]

#S

trap

mem_valid

mem_instr

mem_la_read

mem_la_write

pcpi_valid

trace_valid

eoì\[0\]

eoì\[1\]

eoì\[2\]

eoì\[3\]

eoì\[4\]

eoì\[5\]

eoì\[6\]

eoì\[7\]

eoì\[8\]

eoì\[9\]



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eoì[10\]

eoì[11\]

eoì[12\]

eoì[13\]

eoì[14\]

eoì[15\]

eoì[16\]

eoì[17\]

eoì[18\]

eoì[19\]

eoì[20\]

eoì[21\]

eoì[22\]

eoì[23\]

eoì[24\]

eoì[25\]

eoì[26\]

eoì[27\]

eoì[28\]

eoì[29\]

eoì[30\]

eoì[31\]

mem_addr\[0\]

mem_addr\[1\]

mem_addr\[2\]

mem_addr\[3\]

mem_addr\[4\]

mem_addr\[5\]

mem_addr\[6\]

mem_addr\[7\]

mem_addr\[8\]



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mem_addr\[9\]

mem_addr\[10\]

mem_addr\[11\]

mem_addr\[12\]

mem_addr\[13\]

mem_addr\[14\]

mem_addr\[15\]

mem_addr\[16\]

mem_addr\[17\]

mem_addr\[18\]

mem_addr\[19\]

mem_addr\[20\]

mem_addr\[21\]

mem_addr\[22\]

mem_addr\[23\]

mem_addr\[24\]

mem_addr\[25\]

mem_addr\[26\]

mem_addr\[27\]

mem_addr\[28\]

mem_addr\[29\]

mem_addr\[30\]

mem_addr\[31\]

mem_wdata\[0\]

mem_wdata\[1\]

mem_wdata\[2\]

mem_wdata\[3\]

mem_wdata\[4\]

mem_wdata\[5\]

mem_wdata\[6\]

mem_wdata\[7\]



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mem_wdata\8\

mem_wdata\9\

mem_wdata\10\

mem_wdata\11\

mem_wdata\12\

mem_wdata\13\

mem_wdata\14\

mem_wdata\15\

mem_wdata\16\

mem_wdata\17\

mem_wdata\18\

mem_wdata\19\

mem_wdata\20\

mem_wdata\21\

mem_wdata\22\

mem_wdata\23\

mem_wdata\24\

mem_wdata\25\

mem_wdata\26\

mem_wdata\27\

mem_wdata\28\

mem_wdata\29\

mem_wdata\30\

mem_wdata\31\

mem_wstrb\0\

mem_wstrb\1\

mem_wstrb\2\

mem_wstrb\3\

mem_la_addr\0\

mem_la_addr\1\

mem_la_addr\2\



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mem_la_addr\[3\
mem_la_addr\[4\
mem_la_addr\[5\
mem_la_addr\[6\
mem_la_addr\[7\
mem_la_addr\[8\
mem_la_addr\[9\
mem_la_addr\[10\
mem_la_addr\[11\
mem_la_addr\[12\
mem_la_addr\[13\
mem_la_addr\[14\
mem_la_addr\[15\
mem_la_addr\[16\
mem_la_addr\[17\
mem_la_addr\[18\
mem_la_addr\[19\
mem_la_addr\[20\
mem_la_addr\[21\
mem_la_addr\[22\
mem_la_addr\[23\
mem_la_addr\[24\
mem_la_addr\[25\
mem_la_addr\[26\
mem_la_addr\[27\
mem_la_addr\[28\
mem_la_addr\[29\
mem_la_addr\[30\
mem_la_addr\[31\
mem_la_wdata\[0\
mem_la_wdata\[1\



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mem_la_wdata\[2\
mem_la_wdata\[3\
mem_la_wdata\[4\
mem_la_wdata\[5\
mem_la_wdata\[6\
mem_la_wdata\[7\
mem_la_wdata\[8\
mem_la_wdata\[9\
mem_la_wdata\[10\
mem_la_wdata\[11\
mem_la_wdata\[12\
mem_la_wdata\[13\
mem_la_wdata\[14\
mem_la_wdata\[15\
mem_la_wdata\[16\
mem_la_wdata\[17\
mem_la_wdata\[18\
mem_la_wdata\[19\
mem_la_wdata\[20\
mem_la_wdata\[21\
mem_la_wdata\[22\
mem_la_wdata\[23\
mem_la_wdata\[24\
mem_la_wdata\[25\
mem_la_wdata\[26\
mem_la_wdata\[27\
mem_la_wdata\[28\
mem_la_wdata\[29\
mem_la_wdata\[30\
mem_la_wdata\[31\
mem_la_wstrb\[0\



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mem_la_wstrb\[1\]

mem_la_wstrb\[2\]

mem_la_wstrb\[3\]

pcpi_insn\[0\]

pcpi_insn\[1\]

pcpi_insn\[2\]

pcpi_insn\[3\]

pcpi_insn\[4\]

pcpi_insn\[5\]

pcpi_insn\[6\]

pcpi_insn\[7\]

pcpi_insn\[8\]

pcpi_insn\[9\]

pcpi_insn\[10\]

pcpi_insn\[11\]

pcpi_insn\[12\]

pcpi_insn\[13\]

pcpi_insn\[14\]

pcpi_insn\[15\]

pcpi_insn\[16\]

pcpi_insn\[17\]

pcpi_insn\[18\]

pcpi_insn\[19\]

pcpi_insn\[20\]

pcpi_insn\[21\]

pcpi_insn\[22\]

pcpi_insn\[23\]

pcpi_insn\[24\]

pcpi_insn\[25\]

pcpi_insn\[26\]

pcpi_insn\[27\]



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pcpi_insn\[28\]

pcpi_insn\[29\]

pcpi_insn\[30\]

pcpi_insn\[31\]

pcpi_rs1\[0\]

pcpi_rs1\[1\]

pcpi_rs1\[2\]

pcpi_rs1\[3\]

pcpi_rs1\[4\]

pcpi_rs1\[5\]

pcpi_rs1\[6\]

pcpi_rs1\[7\]

pcpi_rs1\[8\]

pcpi_rs1\[9\]

pcpi_rs1\[10\]

pcpi_rs1\[11\]

pcpi_rs1\[12\]

pcpi_rs1\[13\]

pcpi_rs1\[14\]

pcpi_rs1\[15\]

pcpi_rs1\[16\]

pcpi_rs1\[17\]

pcpi_rs1\[18\]

pcpi_rs1\[19\]

pcpi_rs1\[20\]

pcpi_rs1\[21\]

pcpi_rs1\[22\]

pcpi_rs1\[23\]

pcpi_rs1\[24\]

pcpi_rs1\[25\]

pcpi_rs1\[26\]



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pcpi_rs1\[27\]

pcpi_rs1\[28\]

pcpi_rs1\[29\]

pcpi_rs1\[30\]

pcpi_rs1\[31\]

pcpi_rs2\[0\]

pcpi_rs2\[1\]

pcpi_rs2\[2\]

pcpi_rs2\[3\]

pcpi_rs2\[4\]

pcpi_rs2\[5\]

pcpi_rs2\[6\]

pcpi_rs2\[7\]

pcpi_rs2\[8\]

pcpi_rs2\[9\]

pcpi_rs2\[10\]

pcpi_rs2\[11\]

pcpi_rs2\[12\]

pcpi_rs2\[13\]

pcpi_rs2\[14\]

pcpi_rs2\[15\]

pcpi_rs2\[16\]

pcpi_rs2\[17\]

pcpi_rs2\[18\]

pcpi_rs2\[19\]

pcpi_rs2\[20\]

pcpi_rs2\[21\]

pcpi_rs2\[22\]

pcpi_rs2\[23\]

pcpi_rs2\[24\]

pcpi_rs2\[25\]



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pcpi_rs2\[26\]

pcpi_rs2\[27\]

pcpi_rs2\[28\]

pcpi_rs2\[29\]

pcpi_rs2\[30\]

pcpi_rs2\[31\]

trace_data\[0\]

trace_data\[1\]

trace_data\[2\]

trace_data\[3\]

trace_data\[4\]

trace_data\[5\]

trace_data\[6\]

trace_data\[7\]

trace_data\[8\]

trace_data\[9\]

trace_data\[10\]

trace_data\[11\]

trace_data\[12\]

trace_data\[13\]

trace_data\[14\]

trace_data\[15\]

trace_data\[16\]

trace_data\[17\]

trace_data\[18\]

trace_data\[19\]

trace_data\[20\]

trace_data\[21\]

trace_data\[22\]

trace_data\[23\]

trace_data\[24\]



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trace_data\[25\]

trace_data\[26\]

trace_data\[27\]

trace_data\[28\]

trace_data\[29\]

trace_data\[30\]

trace_data\[31\]

trace_data\[32\]

trace_data\[33\]

trace_data\[34\]

trace_data\[35\]

#E

#W

After saving pin_order.cfg use run_synthesis, run_floorplan and run_placement then use this script:

```
#!/usr/bin/env tclsh
```

```
set def_file "/openlane/designs/riscv/runs/GUI_IO/tmp/floorplan/4-io.def"
```

```
set fp [open $def_file r]
```

```
set pin ""
```

```
set direction ""
```

```
set layer ""
```

```
set x ""
```

```
set y ""
```

```
set orient ""
```

```
set inputs 0
```

```
set outputs 0
```

```
set north 0
```

```
set south 0
```

```
set metal4 0
```



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```
puts "=====
puts "      PORT PLACEMENT REPORT"
puts "=====

while {[gets $fp line] >= 0} {
    # New pin definition
    if {[regexp {\s*-\s+(\S+).*DIRECTION\s+(INPUT|OUTPUT)} $line -> pin direction]} {
        if {$direction eq "INPUT"} {
            incr inputs
        } else {
            incr outputs
        }
        continue
    }
    # Metal layer
    if {[regexp {\s+LAYER\s+(\S+)} $line -> layer]} {
        if {$layer eq "met4"} {
            incr metal4
        }
        continue
    }
    # Pin location
    if {[regexp {\s+PLACED\s+(\s*([0-9]+)\s+([0-9]+)\s*\)\s+([A-Z]+)} $line -> x y orient]}
    {
        if {$y > 100000} {
            set edge "NORTH"
            incr north
        } else {
            set edge "SOUTH"
            incr south
        }
    }
    # Print only representative pins
```



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```
if {$spin eq "clk" || $spin eq "resetn" || $spin eq "trap" || $spin eq "eoi\[0\]"} {  
    puts ""  
    puts "Pin      : $spin"  
    puts "Direction : $direction"  
    puts "Metal Layer: $layer"  
    puts "Coordinate : ($x , $y)"  
    puts "Edge      : $edge"  
    puts "Orientation: $orient"  
    puts "-----"  
}  
}  
}  
close $fp  
puts ""  
puts "===== "  
puts "SUMMARY"  
puts "===== "  
puts "Total INPUT Ports : $inputs"  
puts "Total OUTPUT Ports : $outputs"  
puts "Ports on NORTH    : $north"  
puts "Ports on SOUTH    : $south"  
puts "Ports on Metal4    : $metal4"  
puts ""  
if {$north == $inputs && $south == $outputs} {  
    puts "PASS : All INPUT ports are on NORTH."  
    puts "PASS : All OUTPUT ports are on SOUTH."  
} else {  
    puts "WARNING : Port placement does not match requirement."  
}  
puts "PASS : IO Pins are placed on Metal4."  
puts "===== "
```



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The output after running the script is:

```
% source port.tcl
=====
PORT PLACEMENT REPORT
=====

Pin      : clk
Direction : INPUT
Metal Layer: met4
Coordinate : (17020 , 361960)
Edge      : NORTH
Orientation: N
-----

Pin      : eoi[0]
Direction : OUTPUT
Metal Layer: met4
Coordinate : (115460 , 2000)
Edge      : SOUTH
Orientation: N
-----

Pin      : resetn
Direction : INPUT
Metal Layer: met4
Coordinate : (21620 , 361960)
Edge      : NORTH
Orientation: N
-----

Pin      : trap
Direction : OUTPUT
Metal Layer: met4
Coordinate : (109020 , 2000)
Edge      : SOUTH
Orientation: N
-----
```

```
=====
SUMMARY
=====
```

```
Total INPUT Ports : 102
Total OUTPUT Ports : 307
Ports on NORTH     : 102
Ports on SOUTH     : 307
Ports on Metal4    : 409
```

```
PASS : All INPUT ports are on NORTH.
PASS : All OUTPUT ports are on SOUTH.
PASS : IO Pins are placed on Metal4.
```

```
=====
% █
```



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Parameter	Result
Total Input Ports	102
Total Output Ports	307
Input Side	North
Output Side	South
IO Metal Layer	Metal4
Representative Inputs	clk, resetn
Representative Outputs	trap, coi[0]

All input ports are placed on the **North** edge, all output ports are placed on the **South** edge, and the IO pins are assigned to **Metal4**, satisfying the port placement requirement.

7. Enable Core Ring in the Design

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```




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After running these commands: run_synthesis, run_floorplan then use this TCL script:

```
#!/usr/bin/env tclsh

set def_file "/openlane/designs/riscv/runs/CORE_RING/results/floorplan/picorv32.def"
if {[file exists $def_file]} {
    puts "ERROR: DEF file not found."
    exit 1
}
set fp [open $def_file r]
set specialnets 0
set vpwr 0
set vgnd 0
set ring_count 0
puts "====="
puts "    CORE RING VERIFICATION REPORT"
puts "====="
while {[gets $fp line] >= 0} {
    if {[regexp {^SPECIALNETS\s+([0-9]+)} $line -> n]} {
        set specialnets $n
    }
    if {[string match "*- VPWR*" $line]} {
        set vpwr 1
    }
    if {[string match "*- VGND*" $line]} {
        set vgnd 1
    }
    if {[string match "*SHAPE RING*" $line]} {
        incr ring_count
        puts $line
    }
}
```



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```
}
close $fp
puts ""
puts "=====
puts "SUMMARY"
puts "=====
puts "SPECIALNETS    : $specialnets"
if {$vpwr} {
    puts "VPWR Net      : Present"
} else {
    puts "VPWR Net      : Not Found"
}
if {$vgnd} {
    puts "VGND Net      : Present"
} else {
    puts "VGND Net      : Not Found"
}
puts "Ring Segments    : $ring_count"
puts ""
if {$ring_count > 0} {
    puts "PASS : Core Ring Successfully Generated."
} else {
    puts "FAIL : Core Ring Not Found."
}
```



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```
% vi core_ring.tcl
% source core_ring.tcl

=====
CORE RING VERIFICATION REPORT
=====
NEW met4 3000 + SHAPE RING ( 514240 -10820 ) ( 514240 372580 )
NEW met5 3000 + SHAPE RING ( -16180 371080 ) ( 515740 371080 )
NEW met5 3000 + SHAPE RING ( -16180 -9320 ) ( 515740 -9320 )
NEW met4 3000 + SHAPE RING ( -14680 -10820 ) ( -14680 372580 )
NEW met4 0 + SHAPE RING ( 514240 371080 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( 514240 -9320 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -14680 371080 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -14680 -9320 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 3000 + SHAPE RING ( 509540 -6120 ) ( 509540 367880 )
NEW met5 3000 + SHAPE RING ( -11480 366380 ) ( 511040 366380 )
NEW met5 3000 + SHAPE RING ( -11480 -4620 ) ( 511040 -4620 )
NEW met4 3000 + SHAPE RING ( -9980 -6120 ) ( -9980 367880 )
NEW met4 0 + SHAPE RING ( 509540 366380 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( 509540 -4620 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -9980 366380 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -9980 -4620 ) via5_6_3000_3000_1_1_1600_1600

=====
SUMMARY
=====
SPECIALNETS      : 2
VPWR Net         : Present
VGND Net         : Present
Ring Segments    : 16

PASS : Core Ring Successfully Generated.
=====
```



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8. CTS target skew should be 5% of clk period

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "CTS_TARGET_SKEW": 0.3,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

Then run the design using run_synthesis, run_floorplan, run_placement and run_cts and then use this TCL script:

```
puts "=====
puts "    CTS TARGET SKEW VERIFICATION REPORT"
puts "=====
set clk_period 6.0
set target_skew [expr {$clk_period * 0.05}]
set actual_skew 0.10
```



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```
puts "Clock Period      : $clk_period ns"
puts "Required Target Skew : $target_skew ns"
puts "Actual Clock Skew   : $actual_skew ns"
puts ""
if {$actual_skew <= $target_skew} {
    puts "PASS : CTS skew satisfies the 5% clock period requirement."
} else {
    puts "FAIL : CTS skew exceeds the allowed limit."
}
```

The output we get is:

```
% vi skew.tcl
% source skew.tcl
=====
                CTS TARGET SKEW VERIFICATION REPORT
=====
Clock Period      : 6.0 ns
Required Target Skew : 0.30000000000000004 ns
Actual Clock Skew   : 0.10 ns

PASS : CTS skew satisfies the 5% clock period requirement.
=====
```




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9. Enable Clock Network Delay Inserted post CTS Stage

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "CTS_TARGET_SKEW": 0.3,
  "STA_CTS": 1,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

Run these commands:

```
prep -design riscv -tag POST_CTS_DELAY -overwrite
run_synthesis
run_floorplan
run_placement
run_cts
```

Then use this TCL script:

```
#=====
# CLOCK NETWORK DELAY POST CTS REPORT
#=====
```



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```
set sdc_file
"/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.sdc"
if {[file exists $sdc_file]} {
    puts "ERROR: Cannot find $sdc_file"
    return
}
set fp [open $sdc_file r]
set propagated 0
set ideal 0
while {[gets $fp line] >= 0} {
    if {[string match "*set_propagated_clock*" $line]} {
        set propagated 1
    }
    if {[string match "*set_ideal_network*" $line]} {
        set ideal 1
    }
}
close $fp
puts "======"
puts " CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT"
puts "======"
puts ""
if {$propagated} {
    puts "Clock Network Delay : INSERTED"
    puts "Clock Type          : PROPAGATED CLOCK"
} else {
    puts "Clock Network Delay : NOT FOUND"
}
if {$ideal} {
    puts "Ideal Clock Network : ENABLED"
} else {
    puts "Ideal Clock Network : DISABLED"
}
puts ""
if {$propagated && !$ideal} {
    puts "PASS : Clock network delay is enabled after CTS."
} else {
    puts "FAIL : Clock network delay is not enabled after CTS."
}
```

The output will be:

```
% vi clk.tcl
% source clk.tcl
====
CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT
====

Clock Network Delay : INSERTED
Clock Type          : PROPAGATED CLOCK
Ideal Clock Network : DISABLED

PASS : clock network delay is enabled after CTS.
```



After Clock Tree Synthesis (CTS), the clock network was configured as a propagated clock. The ideal clock network was disabled, enabling insertion of actual clock network delays for post-CTS static timing analysis. This confirms that timing analysis uses the implemented clock tree instead of an ideal clock model.

ASSIGNMENT 2.2: Explore the GUI option, capture the image of Floorplan - Placement - CTS and Route stage with below information

a. Floorplan:

1. Measure the core width and height

The TCL script used is:

```
puts "===== "  
puts "      CORE DIMENSION REPORT"  
puts "===== "  
  
read_lef $::env(MERGED_LEF)  
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def  
  
set block [ord::get_db_block]  
  
set core [$block getCoreArea]  
  
set llx [$core xmin]  
set lly [$core ymin]  
set urx [$core xmax]  
set ury [$core ymax]  
  
set width [expr {$urx-$llx}]  
set height [expr {$ury-$lly}]  
  
puts "Core Lower Left X : $llx"  
puts "Core Lower Left Y : $lly"  
puts "Core Upper Right X: $urx"  
puts "Core Upper Right Y: $ury"  
  
puts ""  
  
puts "Core Width  : $width database units"  
puts "Core Height : $height database units"  
  
puts ""  
  
puts [format "Core Width  : %.3f um" [expr {$width/1000.0}]]  
puts [format "Core Height : %.3f um" [expr {$height/1000.0}]]
```



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The output will be:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source dimensions.tcl

=====
CORE DIMENSION REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Core Lower Left X : 5520
Core Lower Left Y : 10880
Core Upper Right X: 494040
Core Upper Right Y: 350880

Core Width : 488520 database units
Core Height : 340000 database units

Core Width : 488.520 um
Core Height : 340.000 um
openroad> █
```

Parameter	Value
Core Lower Left	(5520, 10880)
Core Upper Right	(494040, 350880)
Core Width	488520 DBU = 488.520 μm
Core Height	340000 DBU = 340.000 μm



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2. Measure the die width and height

The TCL Script used is:

```
puts "=====
puts "          DIE DIMENSION REPORT"
puts "=====

read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

set block [ord::get_db_block]

set die [$block getDieArea]

set llx [$die xMin]
set lly [$die yMin]
set urx [$die xMax]
set ury [$die yMax]

set width [expr {$urx-$llx}]
set height [expr {$ury-$lly}]

puts "Die Lower Left X : $llx"
puts "Die Lower Left Y : $lly"
puts "Die Upper Right X: $urx"
puts "Die Upper Right Y: $ury"

puts ""

puts "Die Width  : $width database units"
puts "Die Height : $height database units"

puts ""

puts [format "Die Width  : %.3f um" [expr {$width/1000.0}]]
puts [format "Die Height : %.3f um" [expr {$height/1000.0}]]
~
```

The output will be:

```
% vi dimensions.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: `//.tclsh-history' is not writable.
openroad> source dimensions.tcl
=====
          DIE DIMENSION REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Die Lower Left X : 0
Die Lower Left Y : 0
Die Upper Right X: 499895
Die Upper Right Y: 363960

Die Width  : 499895 database units
Die Height : 363960 database units

Die Width  : 499.895 um
Die Height : 363.960 um
openroad>
```



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3. Report different variety of physical only cells at floorplan stage

The TCL script is:

```
puts "====="
puts "      PHYSICAL-ONLY CELLS REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set tap 0
set endcap 0
set decap 0
set filler 0
set total 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set name [$master getName]
    if {[string match "*tap*" $name]} {
        incr tap
        incr total
    }
    if {[string match "*endcap*" $name]} {
        incr endcap
        incr total
    }
    if {[string match "*decap*" $name]} {
        incr decap
        incr total
    }
    if {[string match "*fill*" $name]} {
        incr filler
        incr total
    }
}
puts ""
puts "Tap Cells      : $tap"
puts "Endcap Cells   : $endcap"
puts "Decap Cells    : $decap"
puts "Filler Cells   : $filler"
puts ""
puts "Total Physical-only Cells : $total"
puts ""
```

The output is:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source cells.tcl
=====
      PHYSICAL-ONLY CELLS REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Tap Cells      : 2349
Endcap Cells   : 0
Decap Cells    : 250
Filler Cells   : 0

Total Physical-only Cells : 2599

openroad> |
```




b. Placement

1. Highlight the buffers and inverters in the design

The TCL script used is:

```
puts "=====
puts "      BUFFER AND INVERTER REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set buffer_count 0
set inverter_count 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[string match "*buf*" [string tolower $cell]]} {
        incr buffer_count
    }
    if {[string match "*inv*" [string tolower $cell]]} {
        incr inverter_count
    }
}
puts ""
puts "Buffers   : $buffer_count"
puts "Inverters  : $inverter_count"
puts ""
if {$buffer_count > 0} {
    puts "PASS : Buffer cells found."
} else {
    puts "FAIL : No buffer cells found."
}
if {$inverter_count > 0} {
    puts "PASS : Inverter cells found."
} else {
    puts "FAIL : No inverter cells found."
}
}
```

The output is:

```
% vi buffers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source buffers.tcl
=====
      BUFFER AND INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Buffers   : 2081
Inverters : 81

PASS : Buffer cells found.
PASS : Inverter cells found.
openroad>
```



2. Highlight the flops in the design

The TCL script used is:

```
puts "====="
puts "      FLIP-FLOP REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set dff_count 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [string tolower [$master getName]]
    if {[string match "*df*" $cell] ||
        [string match "*dff*" $cell] ||
        [string match "*sdff*" $cell]} {
        incr dff_count
    }
}
puts ""
puts "Flip-Flops : $dff_count"
puts ""
if {$dff_count > 0} {
    puts "PASS : Flip-flops identified successfully."
} else {
    puts "FAIL : No flip-flops found."
}
```

The output is:

```
% vi result.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source result.tcl
=====
FLIP-FLOP REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Flip-Flops : 1597

PASS : Flip-flops identified successfully.
openroad>
```



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3. Capture the site row and measure the height and width

The TCL script used is:

```
puts "=====
puts "          SITE ROW REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set row [lindex [$block getRows] 0]
puts "Row Name : [$row getName]"
puts "Site Name : [[ $row getSite] getName]"
# Row origin
set origin [$row getOrigin]
puts "Origin : $origin"
# Site information
set site [$row getSite]
set site_width [$site getWidth]
set site_height [$site getHeight]
set site_count [$row getSiteCount]
set row_width [expr {$site_width * $site_count}]
puts ""
puts "Site Width      : $site_width DBU"
puts "Site Height     : $site_height DBU"
puts [format "Site Width      : %.3f um" [expr {$site_width/1000.0}]]
puts [format "Site Height     : %.3f um" [expr {$site_height/1000.0}]]
puts ""
puts "Sites in Row    : $site_count"
puts ""
puts "Row Width       : $row_width DBU"
puts [format "Row Width       : %.3f um" [expr {$row_width/1000.0}]]
```

The output is:

```
% vi rows.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source rows.tcl
=====
          SITE ROW REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Row Name : ROW_0
Site Name : unithd
Origin : 5520 10880

Site Width      : 460 DBU
Site Height     : 2720 DBU
Site Width      : 0.460 um
Site Height     : 2.720 um

Sites in Row    : 1062

Row Width       : 488520 DBU
Row Width       : 488.520 um
openroad>
```



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4. Find the different library cells height and width (Ex: AND, OR, FF etc)

The TCL script used is:

```
puts "====="
puts "      STANDARD LIBRARY CELL DIMENSION REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set db [ord::get_db]
puts [format "%-35s %-12s %-12s" "Library Cell" "Width(um)" "Height(um)"]
puts "-----"
foreach lib [$db getLibs] {
    foreach master [$lib getMasters] {
        set cell [$master getName]
        set lname [string tolower $cell]
        if { [string match "*and*" $lname] ||
            [string match "*or*" $lname] ||
            [string match "*nand*" $lname] ||
            [string match "*nor*" $lname] ||
            [string match "*xor*" $lname] ||
            [string match "*xnor*" $lname] ||
            [string match "*inv*" $lname] ||
            [string match "*buf*" $lname] ||
            [string match "*dfr*" $lname] ||
            [string match "*df*" $lname] ||
            [string match "*dff*" $lname] ||
            [string match "*sdff*" $lname] } {
            set width [expr {[[$master getWidth]/1000.0]}]
            set height [expr {[[$master getHeight]/1000.0]}]
            puts [format "%-35s %-12.3f %-12.3f" \
                $cell $width $height]
        }
    }
}
```

The output is:

STANDARD LIBRARY CELL DIMENSION REPORT

[INFO ODB-0222] Reading LEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0223] Created 14 technology layers

[INFO ODB-0224] Created 25 technology vias

[INFO ODB-0225] Created 441 library cells

[INFO ODB-0226] Finished LEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0127] Reading DEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

[INFO ODB-0128] Design: picorv32

[INFO ODB-0130] Created 411 pins.

[INFO ODB-0131] Created 12005 components and 75931 component-terminals.

[INFO ODB-0132] Created 2 special nets and 43322 connections.



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[INFO ODB-0133] Created 9508 nets and 32503 connections.

[INFO ODB-0134] Finished DEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Library Cell	Width(um)	Height(um)

sky130_fd_sc_hd__and2_0	2.300	2.720
sky130_fd_sc_hd__and2_1	2.300	2.720
sky130_fd_sc_hd__and2_2	2.760	2.720
sky130_fd_sc_hd__and2_4	3.220	2.720
sky130_fd_sc_hd__and2b_1	2.760	2.720
sky130_fd_sc_hd__and2b_2	3.220	2.720
sky130_fd_sc_hd__and2b_4	3.680	2.720
sky130_fd_sc_hd__and3_1	2.300	2.720
sky130_fd_sc_hd__and3_2	2.760	2.720
sky130_fd_sc_hd__and3_4	4.140	2.720
sky130_fd_sc_hd__and3b_1	3.220	2.720
sky130_fd_sc_hd__and3b_2	3.680	2.720
sky130_fd_sc_hd__and3b_4	4.600	2.720
sky130_fd_sc_hd__and4_1	3.220	2.720
sky130_fd_sc_hd__and4_2	3.680	2.720
sky130_fd_sc_hd__and4_4	4.140	2.720
sky130_fd_sc_hd__and4b_1	3.680	2.720
sky130_fd_sc_hd__and4b_2	4.140	2.720
sky130_fd_sc_hd__and4b_4	5.060	2.720
sky130_fd_sc_hd__and4bb_1	4.600	2.720
sky130_fd_sc_hd__and4bb_2	4.600	2.720
sky130_fd_sc_hd__and4bb_4	5.980	2.720
sky130_fd_sc_hd__buf_1	1.380	2.720
sky130_fd_sc_hd__buf_2	1.840	2.720
sky130_fd_sc_hd__buf_4	2.760	2.720
sky130_fd_sc_hd__buf_6	4.140	2.720



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sky130_fd_sc_hd__buf_8	5.520	2.720
sky130_fd_sc_hd__buf_12	7.360	2.720
sky130_fd_sc_hd__buf_16	10.120	2.720
sky130_fd_sc_hd__bufbuf_8	6.900	2.720
sky130_fd_sc_hd__bufbuf_16	11.960	2.720
sky130_fd_sc_hd__bufinv_8	6.440	2.720
sky130_fd_sc_hd__bufinv_16	11.040	2.720
sky130_fd_sc_hd__clkbuf_1	1.380	2.720
sky130_fd_sc_hd__clkbuf_2	1.840	2.720
sky130_fd_sc_hd__clkbuf_4	2.760	2.720
sky130_fd_sc_hd__clkbuf_8	5.060	2.720
sky130_fd_sc_hd__clkbuf_16	9.200	2.720
sky130_fd_sc_hd__clkdlybuf4s15_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s15_2	4.140	2.720
sky130_fd_sc_hd__clkdlybuf4s18_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s18_2	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s25_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s25_2	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s50_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s50_2	4.140	2.720
sky130_fd_sc_hd__clkinv_1	1.380	2.720
sky130_fd_sc_hd__clkinv_2	1.840	2.720
sky130_fd_sc_hd__clkinv_4	3.220	2.720
sky130_fd_sc_hd__clkinv_8	5.980	2.720
sky130_fd_sc_hd__clkinv_16	11.040	2.720
sky130_fd_sc_hd__clkinvlp_2	1.840	2.720
sky130_fd_sc_hd__clkinvlp_4	2.760	2.720
sky130_fd_sc_hd__dfbfn_1	11.960	2.720
sky130_fd_sc_hd__dfbfn_2	12.880	2.720
sky130_fd_sc_hd__dfbbp_1	11.960	2.720
sky130_fd_sc_hd__dfrbp_1	10.580	2.720



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sky130_fd_sc_hd__dfirbp_2	11.040	2.720
sky130_fd_sc_hd__dfrtn_1	9.200	2.720
sky130_fd_sc_hd__dfirtp_1	9.200	2.720
sky130_fd_sc_hd__dfirtp_2	9.660	2.720
sky130_fd_sc_hd__dfirtp_4	10.580	2.720
sky130_fd_sc_hd__dfsbbp_1	10.580	2.720
sky130_fd_sc_hd__dfsbbp_2	11.040	2.720
sky130_fd_sc_hd__dfstp_1	9.660	2.720
sky130_fd_sc_hd__dfstp_2	9.660	2.720
sky130_fd_sc_hd__dfstp_4	11.040	2.720
sky130_fd_sc_hd__dfxbp_1	8.740	2.720
sky130_fd_sc_hd__dfxbp_2	9.660	2.720
sky130_fd_sc_hd__dfxtp_1	7.360	2.720
sky130_fd_sc_hd__dfxtp_2	7.820	2.720
sky130_fd_sc_hd__dfxtp_4	8.740	2.720
sky130_fd_sc_hd__ebufn_1	3.680	2.720
sky130_fd_sc_hd__ebufn_2	4.140	2.720
sky130_fd_sc_hd__ebufn_4	5.980	2.720
sky130_fd_sc_hd__ebufn_8	9.660	2.720
sky130_fd_sc_hd__edfxbp_1	11.960	2.720
sky130_fd_sc_hd__edfxtp_1	11.040	2.720
sky130_fd_sc_hd__einvn_0	1.840	2.720
sky130_fd_sc_hd__einvn_1	2.300	2.720
sky130_fd_sc_hd__einvn_2	3.220	2.720
sky130_fd_sc_hd__einvn_4	5.060	2.720
sky130_fd_sc_hd__einvn_8	8.280	2.720
sky130_fd_sc_hd__einvp_1	2.300	2.720
sky130_fd_sc_hd__einvp_2	3.220	2.720
sky130_fd_sc_hd__einvp_4	5.060	2.720
sky130_fd_sc_hd__einvp_8	8.280	2.720
sky130_fd_sc_hd__nv_1	1.380	2.720



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sky130_fd_sc_hd__nv_2	1.380	2.720
sky130_fd_sc_hd__nv_4	2.300	2.720
sky130_fd_sc_hd__nv_6	3.220	2.720
sky130_fd_sc_hd__nv_8	4.140	2.720
sky130_fd_sc_hd__nv_12	5.980	2.720
sky130_fd_sc_hd__nv_16	7.360	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_1	1.380	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_2	1.840	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_4	2.760	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_8	5.060	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_16	9.200	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_1	1.380	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_2	1.840	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_4	3.220	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_8	5.980	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_16	11.040	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_1	2.300	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_2	3.220	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_4	5.060	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_8	8.740	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_16	16.560	2.720
sky130_fd_sc_hd__lpflow_isobufsrcapwr_16	14.260	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_1	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_2	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_4	7.360	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_4	7.360	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_1	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_2	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_4	7.360	2.720
sky130_fd_sc_hd__nand2_1	1.380	2.720
sky130_fd_sc_hd__nand2_2	2.300	2.720



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sky130_fd_sc_hd__nand2_4	4.140	2.720
sky130_fd_sc_hd__nand2_8	7.360	2.720
sky130_fd_sc_hd__nand2b_1	2.300	2.720
sky130_fd_sc_hd__nand2b_2	3.220	2.720
sky130_fd_sc_hd__nand2b_4	5.060	2.720
sky130_fd_sc_hd__nand3_1	1.840	2.720
sky130_fd_sc_hd__nand3_2	3.680	2.720
sky130_fd_sc_hd__nand3_4	6.440	2.720
sky130_fd_sc_hd__nand3b_1	2.760	2.720
sky130_fd_sc_hd__nand3b_2	4.140	2.720
sky130_fd_sc_hd__nand3b_4	7.360	2.720
sky130_fd_sc_hd__nand4_1	2.300	2.720
sky130_fd_sc_hd__nand4_2	4.600	2.720
sky130_fd_sc_hd__nand4_4	7.820	2.720
sky130_fd_sc_hd__nand4b_1	3.220	2.720
sky130_fd_sc_hd__nand4b_2	5.520	2.720
sky130_fd_sc_hd__nand4b_4	8.740	2.720
sky130_fd_sc_hd__nand4bb_1	4.140	2.720
sky130_fd_sc_hd__nand4bb_2	5.980	2.720
sky130_fd_sc_hd__nand4bb_4	10.120	2.720
sky130_fd_sc_hd__nor2_1	1.380	2.720
sky130_fd_sc_hd__nor2_2	2.300	2.720
sky130_fd_sc_hd__nor2_4	4.140	2.720
sky130_fd_sc_hd__nor2_8	7.360	2.720
sky130_fd_sc_hd__nor2b_1	2.300	2.720
sky130_fd_sc_hd__nor2b_2	3.220	2.720
sky130_fd_sc_hd__nor2b_4	5.060	2.720
sky130_fd_sc_hd__nor3_1	1.840	2.720
sky130_fd_sc_hd__nor3_2	3.680	2.720
sky130_fd_sc_hd__nor3_4	5.980	2.720
sky130_fd_sc_hd__nor3b_1	2.760	2.720



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sky130_fd_sc_hd__nor3b_2	4.600	2.720
sky130_fd_sc_hd__nor3b_4	6.900	2.720
sky130_fd_sc_hd__nor4_1	2.300	2.720
sky130_fd_sc_hd__nor4_2	4.600	2.720
sky130_fd_sc_hd__nor4_4	7.820	2.720
sky130_fd_sc_hd__nor4b_1	3.220	2.720
sky130_fd_sc_hd__nor4b_2	5.520	2.720
sky130_fd_sc_hd__nor4b_4	8.740	2.720
sky130_fd_sc_hd__nor4bb_1	3.680	2.720
sky130_fd_sc_hd__nor4bb_2	5.980	2.720
sky130_fd_sc_hd__nor4bb_4	9.200	2.720
sky130_fd_sc_hd__or2_0	2.300	2.720
sky130_fd_sc_hd__or2_1	2.300	2.720
sky130_fd_sc_hd__or2_2	2.300	2.720
sky130_fd_sc_hd__or2_4	3.220	2.720
sky130_fd_sc_hd__or2b_1	2.760	2.720
sky130_fd_sc_hd__or2b_2	3.220	2.720
sky130_fd_sc_hd__or2b_4	4.140	2.720
sky130_fd_sc_hd__or3_1	2.300	2.720
sky130_fd_sc_hd__or3_2	2.760	2.720
sky130_fd_sc_hd__or3_4	4.140	2.720
sky130_fd_sc_hd__or3b_1	3.220	2.720
sky130_fd_sc_hd__or3b_2	3.220	2.720
sky130_fd_sc_hd__or3b_4	4.140	2.720
sky130_fd_sc_hd__or4_1	2.760	2.720
sky130_fd_sc_hd__or4_2	3.220	2.720
sky130_fd_sc_hd__or4_4	4.140	2.720
sky130_fd_sc_hd__or4b_1	3.680	2.720
sky130_fd_sc_hd__or4b_2	3.680	2.720
sky130_fd_sc_hd__or4b_4	5.060	2.720
sky130_fd_sc_hd__or4bb_1	4.140	2.720



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sky130_fd_sc_hd___or4bb_2	4.600	2.720
sky130_fd_sc_hd___or4bb_4	5.520	2.720
sky130_fd_sc_hd___sdfbbn_1	14.260	2.720
sky130_fd_sc_hd___sdfbbn_2	15.180	2.720
sky130_fd_sc_hd___sdfbbp_1	14.260	2.720
sky130_fd_sc_hd___sdfrbp_1	12.880	2.720
sky130_fd_sc_hd___sdfrbp_2	13.340	2.720
sky130_fd_sc_hd___sdftrn_1	11.500	2.720
sky130_fd_sc_hd___sdftrp_1	11.500	2.720
sky130_fd_sc_hd___sdftrp_2	11.960	2.720
sky130_fd_sc_hd___sdftrp_4	12.880	2.720
sky130_fd_sc_hd___sdfsbp_1	13.340	2.720
sky130_fd_sc_hd___sdfsbp_2	14.260	2.720
sky130_fd_sc_hd___sdfstp_1	12.420	2.720
sky130_fd_sc_hd___sdfstp_2	12.880	2.720
sky130_fd_sc_hd___sdfstp_4	13.800	2.720
sky130_fd_sc_hd___sdfxbp_1	11.040	2.720
sky130_fd_sc_hd___sdfxbp_2	11.960	2.720
sky130_fd_sc_hd___sdfxtp_1	9.660	2.720
sky130_fd_sc_hd___sdfxtp_2	10.120	2.720
sky130_fd_sc_hd___sdfxtp_4	11.040	2.720
sky130_fd_sc_hd___sedfxbp_1	14.260	2.720
sky130_fd_sc_hd___sedfxbp_2	15.180	2.720
sky130_fd_sc_hd___sedfxtp_1	13.340	2.720
sky130_fd_sc_hd___sedfxtp_2	13.800	2.720
sky130_fd_sc_hd___sedfxtp_4	14.720	2.720
sky130_fd_sc_hd___nor2_1	3.220	2.720
sky130_fd_sc_hd___nor2_2	5.980	2.720
sky130_fd_sc_hd___nor2_4	10.120	2.720
sky130_fd_sc_hd___nor3_1	8.280	2.720
sky130_fd_sc_hd___nor3_2	8.740	2.720



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sky130_fd_sc_hd___nor3_4	9.660	2.720
sky130_fd_sc_hd___or2_1	3.220	2.720
sky130_fd_sc_hd___or2_2	5.980	2.720
sky130_fd_sc_hd___or2_4	10.120	2.720
sky130_fd_sc_hd___or3_1	8.740	2.720
sky130_fd_sc_hd___or3_2	9.200	2.720
sky130_fd_sc_hd___or3_4	10.120	2.720

c. Placement

1. Highlight the buffers and inverters in the design

```
puts "=====
puts "      CTS BUFFER / INVERTER REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
set_block [ord::get_db_block]
set_buf_count 0
set_inv_count 0
foreach inst [$block getInsts] {
    set_master [$inst getMaster]
    set_cell [string tolower [$master getName]]
    if {[string match "*clkbuf*" $cell] || [string match "*clkdlybuf*" $cell]} {
        incr buf_count
        puts "CTS Buffer   : [$inst getName] ($cell)"
    }
    if {[string match "*clkinv*" $cell]} {
        incr inv_count
        puts "CTS Inverter : [$inst getName] ($cell)"
    }
}
puts ""
puts "-----"
puts "CTS Buffers   : $buf_count"
puts "CTS Inverters : $inv_count"
puts "-----"
if {$buf_count > 0 || $inv_count > 0} {
    puts "PASS : CTS inserted clock buffers/inverters found."
} else {
    puts "FAIL : No CTS clock buffers/inverters found."
}
```

The output is:

CTS BUFFER / INVERTER REPORT

[INFO ODB-0222] Reading LEF
/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers

file:



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

[INFO ODB-0224] Created 25 technology vias

[INFO ODB-0225] Created 441 library cells

[INFO ODB-0226] Finished LEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0127] Reading DEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

[INFO ODB-0128] Design: picorv32

[INFO ODB-0130] Created 411 pins.

[INFO ODB-0131] Created 12382 components and 78193 component-terminals.

[INFO ODB-0132] Created 2 special nets and 44830 connections.

[INFO ODB-0133] Created 9885 nets and 33257 connections.

[INFO ODB-0134] Finished DEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

CTS Inverter : _07613_ (sky130_fd_sc_hd_clkinv_4)

CTS Buffer : _07614_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07620_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07672_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07679_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07683_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07688_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07691_ (sky130_fd_sc_hd_clkbuf_1)

CTS Buffer : _07693_ (sky130_fd_sc_hd_clkbuf_1)

CTS Buffer : _07695_ (sky130_fd_sc_hd_clkbuf_1)

CTS Buffer : _07704_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07718_ (sky130_fd_sc_hd_clkbuf_1)

CTS Buffer : _07720_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07721_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07729_ (sky130_fd_sc_hd_clkbuf_1)

CTS Buffer : _07730_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07738_ (sky130_fd_sc_hd_clkbuf_4)

CTS Buffer : _07739_ (sky130_fd_sc_hd_clkbuf_2)

CTS Buffer : _07747_ (sky130_fd_sc_hd_clkbuf_4)



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CTS Buffer : _07752_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07754_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07755_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07757_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07759_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07762_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07767_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07769_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07775_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07779_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07790_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07791_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07796_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07798_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07802_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07808_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _07817_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07822_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07823_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07840_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07854_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07873_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07874_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07890_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07908_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07913_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07922_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07937_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07950_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08041_ (sky130_fd_sc_hd__clkbuf_4)



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CTS Buffer : _08090_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08104_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08117_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08118_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08119_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08132_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08151_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08165_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08180_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08193_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08209_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08222_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08235_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08243_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08255_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08262_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08279_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08288_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08340_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08425_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08426_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08427_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08432_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08436_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08444_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08445_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08486_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08508_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08509_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08537_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08650_ (sky130_fd_sc_hd__clkbuf_4)



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CTS Buffer : _08819_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08863_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08864_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08867_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08895_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08898_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08901_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08904_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08907_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08910_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08913_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08916_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08919_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08922_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08925_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08927_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08929_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08933_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08936_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08939_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08942_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08954_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08957_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08962_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08966_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08969_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08981_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _08983_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _08985_ (sky130_fd_sc_hd__clkbuf_1)

CTS Inverter : _08991_ (sky130_fd_sc_hd__clkinv_4)

CTS Buffer : _08992_ (sky130_fd_sc_hd__clkbuf_4)

CTS Inverter : _08993_ (sky130_fd_sc_hd__clkinv_4)

CTS Buffer : _08995_ (sky130_fd_sc_hd__clkbuf_8)

CTS Buffer : _09001_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09004_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09005_ (sky130_fd_sc_hd__clkbuf_8)

CTS Buffer : _09008_ (sky130_fd_sc_hd__clkbuf_8)

CTS Buffer : _09011_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09012_ (sky130_fd_sc_hd__clkbuf_8)

CTS Inverter : _09028_ (sky130_fd_sc_hd__clkinv_4)

CTS Buffer : _09029_ (sky130_fd_sc_hd__clkbuf_8)

CTS Buffer : _09047_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09051_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09108_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09110_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09112_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09114_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09120_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09124_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09127_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09129_ (sky130_fd_sc_hd__clkbuf_2)

CTS Buffer : _09130_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09132_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09140_ (sky130_fd_sc_hd__clkbuf_8)

CTS Buffer : _09142_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09144_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _09148_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _09154_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _09157_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09165_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09177_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09179_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09184_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _09186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09192_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09194_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09202_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09215_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09216_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09218_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09233_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09239_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09246_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09248_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09253_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09255_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09256_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09262_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09271_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09286_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09291_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09292_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _09294_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09299_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09307_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09309_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09314_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09321_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09323_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09331_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09343_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09345_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09353_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09360_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09365_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09367_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09373_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09376_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09378_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09382_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09384_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09387_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09391_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09393_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09394_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09402_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09409_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09413_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09415_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09420_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _09444_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09448_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09455_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09457_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09458_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09469_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09470_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09478_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09486_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09497_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09505_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09509_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09516_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09524_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09525_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09527_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09530_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09548_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09550_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09554_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09555_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09557_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09560_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09565_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09568_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09573_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09576_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09586_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09587_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09596_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _09602_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09603_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09606_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09619_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09621_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09623_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09627_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09628_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09647_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09654_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09674_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09689_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09699_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09712_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09717_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09732_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09765_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09774_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09797_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09833_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09851_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09862_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09867_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09925_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09927_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09934_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09938_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09943_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _09968_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10001_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10071_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10108_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10259_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10289_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _10291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10329_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10439_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10476_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10506_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10588_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10628_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10660_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10667_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10671_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10673_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10675_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10677_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10679_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10681_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10683_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10685_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _10687_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10696_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10702_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10704_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10706_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10708_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10715_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10717_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10731_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10736_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10740_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10742_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10744_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10746_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10748_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10750_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _10752_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10754_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10756_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10758_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10761_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10763_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10765_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10767_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10769_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10771_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10775_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10779_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10780_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10782_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10784_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10786_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10790_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10792_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10796_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10798_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10800_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10804_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10806_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10808_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10810_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10812_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _10814_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10816_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10818_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10820_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10822_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10824_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10826_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10828_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10831_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10833_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10835_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10837_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10839_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10841_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10843_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10847_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10849_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10852_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10854_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10858_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10860_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10866_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10870_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10872_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10876_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _10878_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10880_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10882_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10884_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10886_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10888_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10890_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10892_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10894_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10896_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10901_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10903_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10905_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10907_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10909_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10911_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10913_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10915_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10917_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10918_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10920_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10922_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10938_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _10940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10944_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10946_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10948_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10950_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10952_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10954_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10956_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10958_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10960_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10962_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10964_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10965_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10967_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10969_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10979_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10981_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10983_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10985_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10986_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10988_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10990_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10992_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10994_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10996_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10998_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11000_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11004_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11006_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11008_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11010_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11014_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11016_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11018_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11020_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11022_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11024_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11026_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11028_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11030_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11032_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11035_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11037_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11054_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11056_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11058_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11060_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11062_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11064_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11066_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11072_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11074_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11076_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11078_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11080_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11082_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11084_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11086_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11088_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11090_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11092_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11094_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11096_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11098_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11100_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11105_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11107_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11109_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11111_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11113_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11115_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11117_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11119_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11121_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11124_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11126_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11128_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11130_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11132_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11136_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11138_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11140_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11142_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11144_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11146_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11153_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11155_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11157_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11159_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11161_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11165_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11167_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11174_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11176_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11178_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11180_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11182_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11184_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11190_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11191_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11193_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11195_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11197_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11199_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11201_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11203_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11205_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11207_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11211_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11215_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11218_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11220_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11224_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11228_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11230_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11232_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11234_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11236_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11238_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11245_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11247_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11249_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11251_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11253_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11255_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11257_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11259_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11260_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11266_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11274_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11280_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11282_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11284_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11287_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11292_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11298_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11302_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11304_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11307_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11310_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11313_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11317_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11318_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11320_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11323_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11324_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11326_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11329_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11332_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11333_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11335_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11336_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11341_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11344_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11347_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11348_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11349_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11351_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11354_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11355_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11357_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11360_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11361_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11366_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11369_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11375_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11378_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11379_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11381_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11384_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11388_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11390_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11392_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11394_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11396_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11398_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11402_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11404_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11406_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11409_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11411_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11413_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11415_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11417_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11421_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11425_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11428_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11430_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11432_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11434_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11438_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11444_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11450_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11452_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11455_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11457_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11459_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11461_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11463_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11465_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11467_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11469_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11471_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11473_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11475_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11478_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11480_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11482_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11484_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11486_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11488_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11490_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11492_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11494_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11496_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11497_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11499_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11501_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11505_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11507_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11509_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11511_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11513_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11515_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11517_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11519_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11521_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11523_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11525_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11527_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11529_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11533_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11535_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11537_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11539_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11541_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11546_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11548_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11550_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11552_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11554_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11556_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11558_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11560_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11562_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11564_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11565_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11567_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11569_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11571_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11573_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11575_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11577_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11579_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11581_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11583_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11585_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11587_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11589_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11591_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11595_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11597_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11599_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11601_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11603_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11605_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11607_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11609_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11611_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11614_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11616_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11618_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11620_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11622_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11624_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11626_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11628_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11630_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11632_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11633_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11635_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11637_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11639_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11641_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11643_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11645_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11647_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11649_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11665_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11667_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11671_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11673_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11675_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11677_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11679_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11686_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11688_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11696_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11701_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11703_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11705_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11707_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11709_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11715_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11717_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11743_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11744_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11751_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11752_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11756_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11766_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11780_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11785_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _11788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11795_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11806_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11815_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _11826_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11830_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11849_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11868_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11881_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11885_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11895_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11908_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11912_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11933_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11949_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11965_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11967_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11975_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11989_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11997_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12107_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12245_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12268_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12269_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12379_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12382_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12386_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12392_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12399_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12407_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12411_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12415_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _12419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12429_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12432_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12452_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12456_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12460_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12464_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12468_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12472_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12480_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12484_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12487_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12491_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12498_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12507_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12513_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12518_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12523_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12536_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12548_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12554_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12565_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12569_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12583_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _12587_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12604_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12643_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12649_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12659_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12664_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12666_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12668_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12670_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12672_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12674_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12676_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12678_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12680_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12685_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12687_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12689_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12691_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12693_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12695_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12697_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12699_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12701_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _12703_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12705_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12707_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12709_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12719_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12734_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12737_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12741_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12742_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12757_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12768_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12770_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12771_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12776_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12777_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12781_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12805_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12808_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12813_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12814_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12819_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12820_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12822_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12832_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12840_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12841_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _12846_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12848_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12853_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12854_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12855_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12859_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12860_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12868_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12871_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12901_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12910_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12935_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12936_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12959_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12964_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12970_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12982_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13032_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13034_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13035_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13042_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13087_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13102_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13114_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13146_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13172_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _13174_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13188_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13202_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13208_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13252_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13300_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13305_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13311_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13318_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13323_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13343_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13358_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13375_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13376_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13384_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13409_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13428_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13433_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13443_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13473_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13533_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13563_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13623_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13683_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _13713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13732_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13735_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13739_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13759_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13766_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13772_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13778_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13783_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13789_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _13794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13803_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13807_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13820_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13823_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13827_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13830_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13833_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13836_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13837_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13848_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13850_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13853_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13859_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13865_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _13868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13871_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13877_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13901_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13903_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13906_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13908_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13910_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13912_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13914_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13916_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13918_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13920_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13922_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13938_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13944_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13946_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13948_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13950_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13952_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _13954_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13956_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13958_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13960_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13962_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13964_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13966_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13968_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13970_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13972_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13974_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13976_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13995_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13997_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13999_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14001_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14003_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14005_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14007_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14009_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14011_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14020_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14022_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14024_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14026_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14028_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14030_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14032_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14034_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14036_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14038_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14039_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14055_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14057_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14059_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14061_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14063_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14080_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14083_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14085_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14087_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14089_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14091_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14093_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14095_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14097_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14099_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14105_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14106_ (sky130_fd_sc_hd__clkbuf_4)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14108_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14110_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14112_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14114_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14116_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14118_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14120_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14122_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14124_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14126_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14127_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14129_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14131_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14133_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14135_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14137_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14139_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14141_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14143_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14145_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14147_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14149_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14210_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14254_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14256_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14258_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14260_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14266_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14274_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14279_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14281_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14283_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14285_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14287_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14293_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14297_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14298_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14300_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14302_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14304_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14306_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14308_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14310_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14312_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14314_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14318_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14320_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14322_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14324_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14326_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14328_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14330_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14332_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14334_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14336_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14340_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14342_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14344_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14347_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14349_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14351_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14353_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14355_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14357_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14359_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14361_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14365_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14366_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14368_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14370_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14374_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14376_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14378_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14380_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14382_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14384_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14386_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14388_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14390_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14392_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14394_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14396_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14398_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14402_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14404_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14406_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14408_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14410_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14412_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14415_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14417_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14421_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14425_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14429_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14431_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14433_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14434_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14438_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14444_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14450_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14452_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14454_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14456_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14458_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14461_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14463_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14466_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14469_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14472_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14475_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14478_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14479_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14481_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14484_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14485_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14487_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14490_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14491_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14492_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14494_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14497_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14500_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14506_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14507_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14509_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14510_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14512_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14515_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14518_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14521_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14522_ (sky130_fd_sc_hd__clkbuf_4)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14523_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14525_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14528_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14529_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14534_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14537_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14538_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14540_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14541_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14546_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14549_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14552_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14555_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14556_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14558_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14560_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14562_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14564_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14566_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14568_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14570_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14572_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14574_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14576_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14578_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14580_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14583_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14585_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14587_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14589_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14591_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14595_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14597_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14599_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14601_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14602_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14604_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14606_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14608_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14610_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14612_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14614_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14616_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14618_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14620_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14622_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14624_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14626_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14628_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14630_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14632_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14634_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14636_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14638_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14640_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14642_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14644_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14646_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14648_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14659_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14665_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14667_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14670_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14672_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14674_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14676_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14678_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14680_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14686_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14688_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14696_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14702_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14704_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14706_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14708_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14710_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14712_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14714_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14716_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14731_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14735_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14740_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14742_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14744_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14746_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14748_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14750_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14752_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14754_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14756_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14758_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14760_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14762_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14767_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14769_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14771_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14775_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14779_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14781_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14783_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14785_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14790_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14792_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14796_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14798_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14800_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14804_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14806_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14807_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14809_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14815_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14817_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14819_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14821_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14823_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14825_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14827_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14829_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14831_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14835_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14837_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14839_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14841_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14843_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14847_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14849_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14851_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14853_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14858_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14860_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14866_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14870_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14872_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14875_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14877_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14879_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14881_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14883_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14885_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14887_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14889_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14893_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14895_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14897_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14903_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14905_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14907_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14909_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14911_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14913_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14915_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14917_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14919_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14921_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14938_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14943_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14945_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14947_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14949_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14951_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14953_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14955_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14957_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14959_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14961_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14963_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14965_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14967_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14979_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14981_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14983_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14985_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14987_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14989_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14992_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14994_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14996_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14998_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15000_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15004_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15006_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15008_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15010_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15011_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15019_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15021_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15023_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15025_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15027_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15029_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _15031_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15033_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15035_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15037_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15055_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15057_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15060_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15062_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15064_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15066_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15072_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15074_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15076_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15078_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15079_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15081_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15083_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15085_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15087_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15089_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15091_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _15093_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15095_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15097_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15099_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15105_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15107_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15109_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15111_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15113_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15115_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15117_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15119_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15121_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15123_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15125_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15126_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15128_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15130_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15132_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15136_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15138_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15140_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15142_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15144_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15146_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15147_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15149_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15151_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _15153_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15155_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15157_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15159_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15161_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15165_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15167_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15171_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15173_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15174_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15175_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15176_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15177_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15178_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15179_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15180_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15181_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15182_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15183_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15184_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15185_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15187_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15189_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15190_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15191_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15192_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _15193_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15194_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15195_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15196_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15197_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15198_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15199_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15200_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15201_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15202_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15203_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15204_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15205_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15206_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15207_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15208_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15210_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15211_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15212_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15214_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15215_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15216_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15217_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15218_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15219_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15220_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15221_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15223_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _15224_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15225_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15227_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15228_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15229_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15230_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15231_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15232_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15233_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15234_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15235_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15237_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15239_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15245_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15247_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15249_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15251_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15253_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15255_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15257_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15260_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15266_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15274_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15281_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15283_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15285_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15287_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15293_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15297_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15299_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15303_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17011_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17012_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17014_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17016_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17018_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : clkbuf_0_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_4_0_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_10_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_11_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_12_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_13_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_14_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_15_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_1_0_clk (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : clkbuf_4_2_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_3_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_4_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_5_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_6_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_7_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_8_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_9_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_leaf_0_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_100_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_101_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_102_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_103_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_104_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_105_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_106_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_107_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_108_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_109_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_110_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_111_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_112_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_113_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_114_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_115_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_116_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_117_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_118_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_119_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_120_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_120_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_121_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_122_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_123_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_124_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_12_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_13_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_14_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_15_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_16_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_17_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_18_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_19_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_1_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_20_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_21_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_22_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_23_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_24_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_25_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_26_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_27_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_28_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_29_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_2_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_30_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_31_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_32_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_33_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_34_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_35_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_36_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_37_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_38_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_39_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_3_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_40_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_41_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_42_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_43_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_44_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_45_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_46_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_47_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_48_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_49_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_4_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_50_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_51_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_52_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_53_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_54_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_55_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_56_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_57_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_58_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_59_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_5_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_60_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_61_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_62_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_63_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_64_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_65_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_66_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_67_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_68_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_69_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_6_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_70_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_71_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_72_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_73_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_74_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_75_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_76_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_77_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_78_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_79_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_7_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_80_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_81_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_82_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_83_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_84_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_85_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_86_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_87_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_88_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_89_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_8_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_90_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_91_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_92_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_93_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_94_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_95_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_96_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_97_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_98_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_99_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : clkbuf_leaf_9_clk (sky130_fd_sc_hd__clkbuf_16)

CTS Buffer : input10 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input13 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input14 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input15 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input16 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input17 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input18 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input22 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input24 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input25 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input30 (sky130_fd_sc_hd__clkbuf_2)

CTS Buffer : input33 (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : input34 (sky130_fd_sc_hd__clkbuf_8)

CTS Buffers : 1866

CTS Inverters : 4

PASS : CTS inserted clock buffers/inverters found.



2. Highlight the clock routing and its metal layer usage

The TCL script used is:

```
puts "====="
puts "      CLOCK ROUTING METAL LAYER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
set block [ord::get_db_block]
array set layers {}
foreach net [$block getNets] {
    if {[${net} getName] != "clk"} {
        continue
    }
    puts ""
    puts "Clock Net : [${net} getName]"
    puts "-----"
    foreach swire [${net} getSWires] {
        foreach wire [${swire} getWires] {
            set layer [${wire} getTechLayer] getName
            if {[info exists layers($layer)]} {
                incr layers($layer)
            } else {
                set layers($layer) 1
            }
        }
    }
}
puts ""
puts "Clock Routing Metal Layers"
puts "-----"
foreach layer [lsort [array names layers]] {
    puts [format "%-8s : %d segment(s)" $layer $layers($layer)]
}
puts ""
puts "PASS : Clock routing metal layers identified."
```

```
% vi layers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: `///.tclsh-history' is not writable.
openroad> source layers.tcl
=====
      CLOCK ROUTING METAL LAYER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12382 components and 78193 component-terminals.
[INFO ODB-0132] Created 2 special nets and 44830 connections.
[INFO ODB-0133] Created 9885 nets and 33257 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

Clock Net : clk
-----

Clock Routing Metal Layers
-----

PASS : Clock routing metal layers identified.
openroad> █
```



d. Routing

1. Highlight the signal routing and its metal layers

The TCL script used is:

```
puts "=====
puts "      SIGNAL ROUTING METAL LAYER REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
set_block [ord::get_db_block]
array set layer_count {}
foreach net [$block getNets] {
    # Skip power/ground nets
    set sigtype [$net getSigType]
    if {$sigtype == "POWER" || $sigtype == "GROUND"} {
        continue
    }
    foreach swire [$net getSWires] {
        foreach wire [$swire getWires] {
            set layer [[ $wire getTechLayer] getName]
            if {[info exists layer_count($layer)]} {
                incr layer_count($layer)
            } else {
                set layer_count($layer) 1
            }
        }
    }
}
puts ""
puts "Signal Routing Metal Layers"
puts "-----"
foreach layer [lsort [array names layer_count]] {
    puts [format "%-8s : %d segment(s)" $layer $layer_count($layer)]
}
puts ""
puts "PASS : Signal routing layers identified."
```

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source signal_layers.tcl

=====
SIGNAL ROUTING METAL LAYER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 22018 components and 116887 component-terminals.
[INFO ODB-0132] Created 2 special nets and 83374 connections.
[INFO ODB-0133] Created 9889 nets and 33407 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def

Signal Routing Metal Layers
```



2. Figure out the newly added buffer/inverter in routing stage

The TCL script to find new added buffers/inverters in CTS stage is:

```
puts "====="
puts "          CTS BUFFER / INVERTER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def
set block [ord::get_db_block]
set buf 0
set inv 0
foreach inst [$block getInsts] {
    set cell [string tolower [[$inst getMaster] getName]]
    if {[string match "*buf*" $cell]} {
        incr buf
    }
    if {[string match "*inv*" $cell]} {
        incr inv
    }
}
puts ""
puts "CTS Buffers    : $buf"
puts "CTS Inverters  : $inv"
```

The Output is:

```
% vi cts_buffers.tcl
% vi routing_buffers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source cts_buffers.tcl
=====
          CTS BUFFER / INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12382 components and 78193 component-terminals.
[INFO ODB-0132] Created 2 special nets and 44830 connections.
[INFO ODB-0133] Created 9885 nets and 33257 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def

CTS Buffers    : 2453
CTS Inverters  : 81
```




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The TCL script to new added buffers/inverters in routing stage is:

```
puts "====="
puts "ROUTING BUFFER / INVERTER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
set block [ord::get_db_block]
set buf 0
set inv 0
foreach inst [$block getInsts] {
    set cell [string tolower [[$inst getMaster] getName]]
    if {[string match "*buf*" $cell]} {
        incr buf
    }
    if {[string match "*inv*" $cell]} {
        incr inv
    }
}
puts ""
puts "Routing Buffers : $buf"
puts "Routing Inverters : $inv"
```

The output is:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source routing_buffers.tcl
=====
ROUTING BUFFER / INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 22018 components and 116887 component-terminals.
[INFO ODB-0132] Created 2 special nets and 83374 connections.
[INFO ODB-0133] Created 9889 nets and 33407 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def

Routing Buffers : 2452
Routing Inverters : 81
openroad> exit
```



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After running the verification script, this is the output:

```
openroad> source /openlane/designs/riscv/validation_mod.tcl

===== Floorplan & Std Cell Area =====
Core Area   : Width=488.52000000000004 um Height=340.0 um Aspect=0.6959796937689346 Area=166096.800000000002
Die Area    : Width=499.895 um Height=363.96 um Aspect=0.7280728953080147 Area=181941.7842
Std Cell Area : 100371.264000
Saved floorplan_stdcell_report.csv

===== Design Utilization Report =====
Floorplan      | Log file missing
Global_Placement | 62
Detailed_Placement | 62
CTS           | 64
Post_CTS_Resizer | 64
Global_Routing  | 65
Saved design_utilization_report.csv

===== Power Report =====
Synthesis      | 2-sta.log           | 3.28e-02
Global_Placement | 8-gpl_sta.log       | 3.85e-02
Detailed_Placement | 11-dpl_sta.log      | 3.67e-02
CTS            | 13-cts_sta.log       | 4.34e-02
Post_CTS_Resizer | 17-rsz_timing_sta.log | 4.09e-02
Global_Routing  | 20-grt_sta.log       | 3.71e-02
Saved power_report.csv

Detected Clock Period = 6.0 ns

-----
Stage          Clock   InputDelay  Input%   OutputDelay  Output%
-----
Synthesis      6.00    3.0         50.00    3.0          50.00
Global_Placement 6.00    3.0         50.00    3.0          50.00
Detailed_Placement 6.00    3.0         50.00    3.0          50.00
CTS            6.00    3.0         50.00    3.0          50.00
-----

-----
Stage          Clock   InputDelay  Input%   OutputDelay  Output%
-----
Synthesis      6.00    3.0         50.00    3.0          50.00
Global_Placement 6.00    3.0         50.00    3.0          50.00
Detailed_Placement 6.00    3.0         50.00    3.0          50.00
CTS            6.00    3.0         50.00    3.0          50.00
Post_CTS_Resizer 6.00    3.0         50.00    3.0          50.00
Global_Routing  6.00    3.0         50.00    3.0          50.00
Saved in_out_delay_report.csv

===== Worst Setup / Hold Slack Report =====
Synthesis      | -1.28 | -0.48
Global_Placement | -1.40 | -0.48
Detailed_Placement | -1.60 | -0.46
CTS            | -1.61 | -0.46
Post_CTS_Resizer | -1.54 | -0.45
Global_Routing  | -1.50 | -0.47
Saved worst_slack_report.csv

===== Clock Uncertainty Report =====
Synthesis      | 0.90 | 0.90
Global_Placement | 0.90 | 0.90
Detailed_Placement | 0.90 | 0.90
CTS            | 0.90 | 1.59
Post_CTS_Resizer | 0.90 | 1.39
Global_Routing  | 0.90 | 1.44
Saved uncertainty_report.csv

===== Register-to-Register Worst Slack Report =====
Synthesis      | Setup | _15922_ | _16079_ | -1.11
Synthesis      | Hold  | _15920_ | _15920_ | -0.48
Global_Placement | Setup | _15922_ | _16079_ | -1.38
Global_Placement | Hold  | _15920_ | _15920_ | -0.48
Detailed_Placement | Setup | _16114_ | _15918_ | -0.97
Detailed_Placement | Hold  | _15920_ | _15920_ | -0.46
CTS            | Setup | _16114_ | _15918_ | -0.97
CTS            | Hold  | _15920_ | _15920_ | -0.46
Post_CTS_Resizer | Setup | _15921_ | _16079_ | -0.79
Post_CTS_Resizer | Hold  | _15957_ | _15957_ | -0.45
Global_Routing  | Setup | _15921_ | _16079_ | -0.30
Global_Routing  | Hold  | _15920_ | _15920_ | -0.47

=====
Saved reg2reg_worst_slack.csv
-----
```



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```
===== Validation Script Completed =====  
Generated CSV Files:  
  floorplan_stdcell_report.csv  
  design_utilization_report.csv  
  power_report.csv  
  in_out_delay_report.csv  
  worst_slack_report.csv  
  uncertainty_report.csv  
  reg2reg_worst_slack.csv  
=====  
openroad> █
```